

Chapter 2

Background: Quantum Communication and Information Theory

ABSTRACT

In this chapter, we discuss the basic notation and terminology behind quantum computation and communication. We discuss the primary operations allowed within quantum mechanics, and we also introduce several concepts from quantum information theory that will be important later. The reader familiar with basic quantum computation may skip this chapter, referring back to it as needed.

KEYWORDS

Quantum Communication, Quantum Information Theory

In this chapter, we will introduce the basic concepts and notation required to understand the rest of this textbook. Many other valuable resources exist to supplement the reading here. Of particular note is the excellent textbook ?.

Road-map: This chapter begins with a review of some basic linear algebra, in Section 2.1. While this material may be familiar, it is introduced using Dirac notation, which we will use throughout this book, and so it may be helpful to review this section even if you have a working knowledge of the underlying material.

In Section 2.2, we will start introducing basic quantum systems, quantum states, and the concept of superposition. Measurements will be discussed in Section 2.3, including basis measurements, and general measurement operators. Section 2.4 will discuss state evolution via unitary operators while Section 2.5 will discuss tensor product and how we combine multiple quantum states. Entanglement will also be discussed in that section, along with partial measurements.

Finally, section 2.6 discusses quantum information theory. We begin with some basic Shannon information theory. Density operator formalism is discussed in Section 2.6.1, including measurements, state evolution, partial trace, and trace distance. Finally, quantum entropy is discussed in Section 2.6.2, including von Neumann entropy and quantum min entropy.

2.1 BASIC LINEAR ALGEBRA

As it turns out, all the mathematics behind quantum computing and cryptography, from modeling quantum states to arguing about security, has a foundation in linear algebra. So, in this section we'll review some of the basics that you'll need to understand the rest of the text. Readers familiar with linear algebra may consider glossing over this section. We'll be introducing the basic concepts, however, using *Dirac notation* which is the main notation system used in quantum computing research. *Thus, even if you are familiar with basic linear algebra, you may want to at least skim this section to become familiar with the notation we'll be using throughout.* Alternatively, you may want to skip ahead to Section 2.2 and refer back to this section when something doesn't make sense in the notation. This section is not meant to be an exhaustive survey of all linear algebra topics needed, but many of the important issues are covered here.

The most basic element in linear algebra is the *vector*. Abstractly, vectors are objects that can be added together and also multiplied with a scalar. For our purposes, the vectors we care about are n -dimensional complex vectors, that is n -tuples with each element a complex number. We denote the space of such vectors as $\mathbb{C}^n$. A vector will be written using Dirac notation as a *ket* denoted $|\cdot\rangle$, where the $\cdot$ can be any label you like. Let's consider the "simple" case of $n = 2$. A vector, then, can be written as:

$$|\text{my-vector}\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad (2.1)$$

where α and β are two complex numbers (numbers of the form $x + yi$ where $i^2 = -1$). Above, the vector is called "my-vector". Moving forward, we'll be using Greek letters to denote vectors typically, for example $|\psi\rangle$ which is traditional in quantum research - however, feel free to call your vectors whatever you like.

Given two vectors $|\psi\rangle$ and $|\phi\rangle$ one may add them together to form a new vector $|\psi\rangle + |\phi\rangle$. If one writes them as tuples (as in Equation 2.1), then the resulting vector is simply the addition of each coordinate together, namely:

$$\underbrace{\begin{pmatrix} a \\ b \end{pmatrix}}_{|\psi\rangle} + \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{|\phi\rangle} = \underbrace{\begin{pmatrix} a+x \\ b+y \end{pmatrix}}_{|\psi\rangle+|\phi\rangle}. \quad (2.2)$$

The above naturally generalizes to any dimension n . Besides addition, one also has *scalar multiplication*: namely, given a complex number $z \in \mathbb{C}$ and a vector $|\psi\rangle$, we can write $z|\psi\rangle$ to mean:

$$\underbrace{z \begin{pmatrix} a \\ b \end{pmatrix}}_{|\psi\rangle} = \underbrace{\begin{pmatrix} za \\ zb \end{pmatrix}}_{z|\psi\rangle} \quad (2.3)$$

Above, we used both Dirac notation, and also Cartesian-style vectors (e.g., arrays of numbers). From an educational point of view, it is a delicate matter as to how one should exactly view a vector $|\psi\rangle$. In general, we recommend thinking of them as Cartesian vectors only when first learning, but to try very hard to never write out, explicitly, what the vector is as a tuple - instead it is much more elegant, and generally easier, to work directly with Dirac notation and to not use tuples. However, there are times when you will want to work with coordinates/tuples and indeed it may be easier to do so. There is no hard and fast rule - in general working with Dirac notation (i.e., vectors abstractly) is usually easier in most situations, however when first starting, and even for some advanced problems, working with Cartesian coordinate vectors will be easier. Moving forward we'll work in both systems for educational purposes, slowly switching away from Cartesian coordinates as we move on.

Vector arithmetic follows a lot of the properties we're used to working with. In particular, the distributive property holds, namely for any scalar $z \in \mathbb{C}$ and vectors $|u\rangle, |v\rangle$ (of the same dimension), then:

$$z(|u\rangle + |v\rangle) = z|u\rangle + z|v\rangle. \quad (2.4)$$

Other important properties will become apparent as we move forward.

2.1.1 Vector Spaces and Bases

One of the most important concepts in linear algebra, in particular when working with vectors, is the *vector space*. A vector space V over $\mathbb{C}$ is a set of *vectors* such that certain properties are satisfied. Of importance is that the vector space is closed under addition and scalar multiplication. This means for any two vectors $|u\rangle$ and $|v\rangle$ from the set V , if we add them together to form a new vector $|w\rangle = |u\rangle + |v\rangle$, it holds that $|w\rangle \in V$. Furthermore, if we take any $\alpha \in \mathbb{C}$, then the product $\alpha|u\rangle$ also lives in V .

One example of a vector space is simply $\mathbb{C}^n$. Indeed, take any two vectors from this space and add them together, you will end up with an n -dimensional vector in $\mathbb{C}^n$; similarly if you were to multiply with any complex number α . A slightly more complicated example (though only slightly so) would be the vector space $V = \{(\alpha, 0)^T : \alpha \in \mathbb{C}\}$. Note that, here, we use v^T to mean the transpose of v , while later we will use v^* to mean the conjugate transpose.

Clearly, the above is a *vector subspace* of $\mathbb{C}^2$. That it is closed under addition and scalar multiplication is straight-forward to see. An example of a set that is *not* a vector space would be $\{(\alpha, 0) : \alpha \text{ is an even integer}\}$. While this is closed under addition, it is not closed under scalar multiplication (just multiply by $\frac{1}{2}$). As we progress, we will encounter varying vector spaces and subspaces, though in this text they will all ultimately be subspaces of $\mathbb{C}^n$ for some n .

Now, consider a vector space V (which we may think of as a subspace of $\mathbb{C}^n$). A *spanning set* is a set $S \subset V$ such that for any $|v\rangle \in V$, we can write $|v\rangle$ as a

linear combination of vectors in $\mathcal{S}$. Namely, we can find scalars such that:

$$|v\rangle = \sum_{|s\rangle \in \mathcal{S}} \alpha_s |s\rangle \quad (2.5)$$

Given such a set, we say $\mathcal{S}$ *spans* V .

A *basis* is a set $\mathcal{B} = \{|b_1\rangle, |b_2\rangle, \dots\} \subset V$ that spans V , and such that no vector in $\mathcal{B}$ may be written as a linear combination of other vectors in $\mathcal{B}$. That is, for every $|b_i\rangle \in \mathcal{B}$, we cannot find scalars α_j such that:

$$|b_i\rangle = \sum_{j \neq i} \alpha_j |b_j\rangle \quad (2.6)$$

In a way, a basis is a minimal spanning set. Namely, start with any spanning set. If there exists a vector in the set that can be written as a linear combination of the others, then one may discard it (it is, essentially, redundant, as it can already be written as other vectors in the set).

As an example, the so-called *standard basis* of $\mathbb{C}^n$ is $\{(1, 0, 0, \dots, 0)^T, (0, 1, 0, \dots, 0)^T, \dots, (0, 0, 0, \dots, 1)^T\}$. That any vector $|v\rangle \in \mathbb{C}^n$ can be written as a linear combination of vectors from this set, is trivial. It is also clear that no vector in this basis can be written as a linear combination of the others. Note that there are n vectors in this set. This is no coincidence. In fact, we define the *dimension* of a vector space V to be the number of vectors in a basis of that space. While there may be infinitely many bases for any one vector space, they will always contain the same number of vectors!

Exercise 2.1. Consider the following problems:

1. Show that $\{(1, 1)^T, (1, -1)^T\}$ is a basis for $\mathbb{C}^2$.
2. Show that $\{(2, 0)^T\}$ is a basis for the vector subspace $V = \{(\alpha, 0)^T : \alpha \in \mathbb{C}\}$ (which was discussed earlier in this subsection).

Bases are important building blocks of vector spaces; in a way they are the “atoms” of a vector space. They will be very important later, when we discuss quantum states and measurements.

2.1.2 Inner Products

Let V be a vector space. We say V is an *inner product space* if there exists a function $f : V \times V \rightarrow \mathbb{C}$ (i.e., a function that takes two vectors as inputs and outputs a complex number) which satisfies the following properties:

1. The function must be linear in its second argument, namely: $f(|v\rangle, \sum_i \alpha_i |w_i\rangle) = \sum_i \alpha_i f(|v\rangle, |w_i\rangle)$.
2. If you reverse the order of the inputs, the outputs are complex conjugates of one another; namely $f(|v\rangle, |w\rangle) = (f(|w\rangle, |v\rangle))^*$.
3. Finally, the inner product of a vector with itself is always a real, non-negative number, namely $f(|v\rangle, |v\rangle) \geq 0$, for any vector $|v\rangle$; equality holds, only if $|v\rangle$ is the zero vector.

We are dealing with vectors involving complex numbers, namely $\mathbb{C}^n$ (or some subspace of $\mathbb{C}^n$). In this case, we may use the standard *dot-product* as our inner product function. Given $|v\rangle = (\alpha_1, \dots, \alpha_n)^T$ and $|w\rangle = (\beta_1, \dots, \beta_n)^T$, then the inner product is $f(|v\rangle, |w\rangle) = \sum_i \alpha_i^* \beta_i$. The reader may readily verify that this definition satisfies the above three necessary properties.

The notation $f(\cdot, \cdot)$ is non-standard and, in fact, rather “clunky.” Dirac notation admits a beautiful simplification for handling inner product. Given a vector $|v\rangle = (\alpha_1, \dots, \alpha_n)^T$, we define by $\langle v|$ (read “bra v ”), to be the conjugate transpose of $|v\rangle$. Namely:

$$|v\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \Rightarrow \langle v| = (\alpha_1^*, \dots, \alpha_n^*). \quad (2.7)$$

Note that, if using Cartesian vectors, then $\langle v|$ is a row vector, while $|v\rangle$ is a column vector. Consider, now, the matrix product $\langle v| \times |w\rangle$ (for some other n -dimensional vector $|w\rangle = (\beta_1, \dots, \beta_n)^T$). This works out to be a 1×1 matrix which, of course, we may take to be simply a scalar value. The product is exactly the inner product:

$$\langle v| \times |w\rangle = (\alpha_1^*, \dots, \alpha_n^*) \times \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix} = \sum_i \alpha_i^* \beta_i. \quad (2.8)$$

Inner products are such a common operation, we will compress the notation and write, simply $\langle v|w\rangle$. This is a “braket” (a play on the word bracket, hence the name for this Dirac “BraKet” notation).

For computing inner products, one may mentally switch back and forth between thinking of a braket as an inner product and thinking of it as the multiplication of two vectors (one of which you must take the conjugate transpose first). For example, let’s define:

$$|+\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle, \quad (2.9)$$

(the above is an important vector that will show up often); and now, let’s compute $\langle 0|+\rangle$:

$$\langle 0|+\rangle = \langle 0| \times \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) = \frac{1}{\sqrt{2}} \langle 0|0\rangle + \frac{1}{\sqrt{2}} \langle 0|1\rangle. \quad (2.10)$$

If you recall that we define $|0\rangle$ and $|1\rangle$, typically, as the standard basis, one may readily verify that $\langle 0|0\rangle = 1$ and $\langle 0|1\rangle = 0$. Thus:

$$\langle 0|+\rangle = \frac{1}{\sqrt{2}}. \quad (2.11)$$

Above, we took advantage of the linearity of matrix multiplication; however we could also have taken advantage of linearity of inner product for the above computation. In general, when one sees a bracket, you should think inner product; but if it's easier to compute something by thinking of it as a matrix multiplication, go ahead and do so. The very curious reader may want to read more of the theory behind this notation and the reasoning behind it, for that we recommend ?.

Now, above, we said that the inner product of $|0\rangle$ and $|1\rangle$ were zero, but this assumed we used the standard basis $|0\rangle = (1, 0)^T$ and $|1\rangle = (0, 1)^T$. What if we were to use other vector representations for these states? Well, as it turns out, we will from now on simply define the states $|0\rangle$ and $|1\rangle$ to be *orthonormal*. Namely, they will always be *orthogonal* and they will always be *normalized*:

Definition 2.1.1. Two vectors $|x\rangle$ and $|y\rangle$ are *orthogonal* if their inner product is zero, namely if:

$$\langle x|y\rangle = \langle y|x\rangle = 0. \quad (2.12)$$

A vector $|w\rangle$ is *normalized* if its inner product is one, namely:

$$\langle w|w\rangle = 1. \quad (2.13)$$

Given any vector $|v\rangle$ (which may not be normalized), one may always *normalize it* by dividing by the square root of the inner product of itself. Namely, the normalized form of $|v\rangle$ is:

$$\frac{|v\rangle}{\sqrt{\langle v|v\rangle}}. \quad (2.14)$$

Note that if $|v\rangle$ is already normalized, then the normalized form is identical to $|v\rangle$. Otherwise, the new vector will always have inner product one with itself.

Returning to our $|0\rangle$ and $|1\rangle$ states, if we define them as orthonormal, then it won't matter what tuple vector representation we use, we will always know that each one is normalized and orthogonal. If we then define other vectors with respect to these, we will always be able to compute inner products of those new vectors also.

Exercise 2.2. Let $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Compute $\langle +|+\rangle$, $\langle +|-\rangle$, and $\langle -|-\rangle$. What can you say about these two vectors? You may use the fact that $|0\rangle$ and $|1\rangle$ are orthonormal.

While we discussed general bases above, one very important type of basis used in quantum computing and cryptography will be *orthonormal bases*. Let V be an inner product space (i.e., a vector space with an inner product) and let $\mathcal{B} = \{|b_1\rangle, |b_2\rangle, \dots, |b_n\rangle\}$ be a subset of vectors. Then we say $\mathcal{B}$ is an *orthonormal basis of V* if:

1. $\mathcal{B}$ is a basis of V .
2. Each vector is pair-wise orthogonal. Namely, for every $i \neq j$, it holds that $\langle b_i|b_j\rangle = 0$.

3. Each vector is normalized. Namely, for each i , it holds that $\langle b_i | b_i \rangle = 1$.

Note that, conditions two and three, above, can be combined by writing:

$$\langle b_i | b_j \rangle = \delta_{i,j}, \quad (2.15)$$

where $\delta_{i,j}$ is the Kronecker delta function; namely $\delta_{i,j} = 1$, only if $i = j$ and is zero otherwise.

Before leaving, we'll mention one important inequality involving inner product, namely the Cauchy-Schwarz inequality. Let $|v\rangle$ and $|w\rangle$ be two vectors, then it holds that:

$$|\langle v | w \rangle| \leq \sqrt{\langle v | v \rangle \langle w | w \rangle}. \quad (2.16)$$

This will be useful in many situations in this book.

2.1.3 Linear Operators

Let V and W be two vector spaces. A *linear operator* is a function A mapping vectors from V to vectors in W , namely $A : V \rightarrow W$, that is also linear in the sense that:

$$A \left(\sum_i \alpha_i |v_i\rangle \right) = \sum_i \alpha_i A(|v_i\rangle). \quad (2.17)$$

To avoid overly cumbersome notation, we will write $A|v\rangle$ instead of $A(|v\rangle)$. Thus, the above, becomes simply:

$$A \sum_i \alpha_i |v_i\rangle = \sum_i \alpha_i A|v_i\rangle. \quad (2.18)$$

One commonly understood case of a linear operator is a matrix and the use of matrix multiplication. Namely, if A is an $m \times n$ matrix, then A will map (column) vectors from $\mathbb{C}^n$ to vectors in $\mathbb{C}^m$. However, representing A as a matrix, is only one way to represent a linear operator. And, while in our case, matrix representations are not missing anything (i.e., all linearity operators from $\mathbb{C}^n$ to $\mathbb{C}^m$ can be represented as a matrix), such representations become cumbersome as the dimension of the systems involved increase.

Instead of matrix representations, we will often represent linear operators by their action on basis states. Note that, due to linearity, if we explicitly define the action of A , our operator, on basis vectors, then we will know exactly how A behaves on *any* vector. Thus, if $\{|b_1\rangle, \dots, |b_n\rangle\}$ is a basis, and if we know what $A|b_i\rangle$ is, for every i , then we will know exactly how A behaves on $|\psi\rangle$ since we can always write $|\psi\rangle = \sum_i \alpha_i |b_i\rangle$, for some scalars α_i . The action will simply be:

$$A|\psi\rangle = \sum_i \alpha_i A|b_i\rangle. \quad (2.19)$$

Exercise 2.3. Let Z be a linear operator defined by:

$$\begin{aligned} Z|0\rangle &= |0\rangle \\ Z|1\rangle &= -|1\rangle. \end{aligned}$$

Then, what is $Z|+\rangle$ and $Z|-\rangle$ (where $|\pm\rangle$ were defined in Exercise 2.2). Write out an equivalent matrix representation of Z :

$$Z = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \quad (2.20)$$

(i.e., fill in numbers for a, b, c , and d)

Another way to represent an operator is through *outer product representation*. Let $|v\rangle$ be a vector in some vector space V , and $|w\rangle$ be a vector in vector space W ; we define a linear operator $|w\rangle\langle v|$ which maps vectors from V to W via the following action:

$$|w\rangle\langle v| |\psi\rangle = \langle v|\psi\rangle |w\rangle. \quad (2.21)$$

That is, on input some vector $|\psi\rangle \in V$, the output will be $|w\rangle \in W$, with a scalar coefficient of the inner product $\langle v|\psi\rangle$. Note that, again, Dirac notation shines - if you forget that the meaning of $|w\rangle\langle v|$ is a linear operator with the action discussed above, then, if you are ever faced with computing $|w\rangle\langle v| |\psi\rangle$, just think of it as vector multiplication (where, recall $\langle v|$ is the conjugate transpose of $|v\rangle$)! In this case: $|w\rangle \times \langle v| \times |\psi\rangle = |w\rangle (\langle v|\psi\rangle) = \langle v|\psi\rangle |w\rangle$, where, for the last equality, we took advantage of the fact that $\langle v|\psi\rangle$ is a scalar. In general, then, a linear operator may be defined by summing over multiple operators of the form $\alpha_{v,w} |w\rangle\langle v|$ (for different vectors $|v\rangle$ and $|w\rangle$ and a scalar $\alpha_{v,w}$).

Given a vector space V , one very important, and at the same time trivial, operator is the *identity operator* denoted I_V (or, simply, I if the context is clear). This operator simply maps any vector to itself, namely $I|\psi\rangle = |\psi\rangle$ for all $|\psi\rangle \in V$. Let $\{|b_1\rangle, \dots, |b_n\rangle\}$ be an orthonormal basis and define the operator $U = \sum_i |b_i\rangle\langle b_i|$. Then, it turns out, $U = I$ (i.e., U is the identity operator). This is known as the *completeness relation* and will be an important tool in analyzing and proving results later. To see why U is the identity operator, first note that $U|b_i\rangle = |b_i\rangle$. This can be seen using Equation 2.21:

$$U|b_i\rangle = \left(\sum_j |b_j\rangle\langle b_j| \right) |b_i\rangle = \sum_j |b_j\rangle\langle b_j|b_i\rangle = \sum_j \langle b_i|b_j\rangle |b_j\rangle = |b_i\rangle. \quad (2.22)$$

The last equality follows from the fact that, since the basis vectors are orthonormal, $\langle b_i|b_j\rangle = 0$ whenever $i \neq j$ (and is one otherwise). Now, let $|\psi\rangle$ be any state. Since $\{|b_i\rangle\}$ for a basis, we can write $|\psi\rangle = \sum_i \alpha_i |b_i\rangle$. Then:

$$U|\psi\rangle = \sum_i \alpha_i U|b_i\rangle = \sum_i \alpha_i |b_i\rangle = |\psi\rangle. \quad (2.23)$$

Since $|\psi\rangle$ was arbitrary, the result is proven.

In summary, we have three ways to represent a linear operator: as a matrix, by its action on basis states, and by its outer-product form. Any of the three may be used; depending on the application, usually one method is easier, or more natural, than another. The reader will encounter each of the methods throughout this book.

We'll close this section by defining certain types of special operators that will be important later. One important type of linear operator is known as a *unitary operator*:

Definition 2.1.2. A *unitary operator* U acting on some complex vector space V (which we will assume is finite dimensional here), is an operator such that:

$$UU^* = U^*U = I, \quad (2.24)$$

where I is the identity operator and U^* is the conjugate transpose of U .

Another important type of linear operator is known as a *Hermitian operator*:

Definition 2.1.3. An operator H is *Hermitian* if $H^* = H$, where H^* is the conjugate transpose of H (the adjoint operator).

Definition 2.1.4. A *positive operator*, A , is one where, for any vector $|\zeta\rangle$, it holds that:

$$\langle \zeta | A | \zeta \rangle > 0. \quad (2.25)$$

A *positive semi-definite* operator is the same, but where $\langle \zeta | A | \zeta \rangle \geq 0$.

2.1.4 Matrix Functions

Let A be a linear operator. Then, an *eigenvector* is a non-zero vector $|\phi\rangle$ such that $A|\phi\rangle = \lambda|\phi\rangle$, where $\lambda \in \mathbb{C}$ is the corresponding *eigenvalue*. An operator A is said to be *diagonalizable* if it can be written as:

$$A = \sum_i \lambda_i |\phi_i\rangle \langle \phi_i|, \quad (2.26)$$

where $|\phi_i\rangle$ are orthonormal eigenvectors with corresponding eigenvalues λ_i . Note that any *normal operator* (i.e., one where $AA^* = A^*A$) is diagonalizable.

Now, given some function $f : \mathbb{C} \rightarrow \mathbb{C}$, and some diagonalizable operator A , we can define a matrix function $f(A)$ by extending the function f to work on operators. The definition will simply be:

$$f(A) = \sum_i f(\lambda_i) |\phi_i\rangle \langle \phi_i|, \quad (2.27)$$

where, again, λ_i and $|\phi_i\rangle$ are the eigenvalues and orthonormal eigenvectors of operator A . For example, consider:

$$Z = 1 \cdot |0\rangle \langle 0| + (-1) \cdot |1\rangle \langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (2.28)$$

Then, consider the function $f(x) = e^{\pi x}$ and, so, $f(Z) = e^{\pi Z}$ is defined to be:

$$e^{\pi Z} = e^{\pi} |0\rangle \langle 0| + e^{-\pi} |1\rangle \langle 1| = \begin{pmatrix} e^{\pi} & 0 \\ 0 & e^{-\pi} \end{pmatrix}. \quad (2.29)$$

2.2 QUANTUM STATES

Quantum states in a closed system are modeled as normalized vectors in some complex vector space, denoted $|\cdot\rangle$ using Dirac notation, where the “.” can be any label. To be more precise, they are elements of a complex Hilbert space - however, since all the states we deal with in this text are finite dimensional, we can restrict our definitions to vectors in $\mathbb{C}^n$ for some n . The space our state lives in is generally called the *state space* of the system, while the actual vector is called the *state*, or *state vector*. Normalized, recall, implies that $\langle\psi|\psi\rangle = 1$. This is often known as the first postulate of quantum computing ?. By “closed system” we mean the state is completely self-contained and not interacting with some outside system. Later, we will explore open quantum systems - vital for modeling quantum communication and cryptographic protocols.

Quantum States: A closed quantum system can be modeled as a normalized vector (called the *state*) living in some Hilbert space (called the *state space*). In this text, all systems are finite dimensional, so the state space is $\mathbb{C}^n$ for some n (the dimension of the state).

A *qubit* or *quantum bit* is a two-dimensional quantum state, i.e., some state $|\psi\rangle \in \mathbb{C}^2$. Qubits are really just mathematical abstractions of some physical particle. This is not dissimilar from a classical bit (a 0 or 1) which is also a mathematical abstraction; indeed, when we discuss algorithms and protocols in classical computer science, we often don’t care too much about *how* bits are implemented only that they can be. Of course, this is not always the case - when one starts investigating hardware issues of computers, then one is very interested in the physical implementation. However, bits, by themselves, are abstract place holders for some physical thing: a thing that can be in a “high” or “low” state. There are many ways to implement a bit: for example a high or low voltage signal; a charged or discharged capacitor; or a full or empty glass of water. When designing an algorithm or cryptographic protocol, mostly one does not actually care about the exact physical implementation and, so, one usually just abstractly works with bits.

The thinking is similar in quantum computing: a qubit is just our abstraction of some physical object; however just like there are several ways to physically implement a classical bit, there are several ways to physically implement a qubit. Normally, when designing and analyzing algorithms and cryptographic protocols at a high-level, we don’t care too much how they’re implemented. This, of course, is not a general rule and there are often times where you need to consider the physical implementation. Nonetheless, for the sake of our discussion in this text, qubits will remain abstract, and we will only refer to specific physical

implementations when needed or if it helps with the understanding.

Let's build a better understanding, now, of qubits and their many differences to classical bits (c-bits). One of the very first differences comes in the number of possible states a c-bit versus a qubit may exist in. A c-bit can be in only one of two possible states at a single time: it is either zero or it's one. A single qubit, on the other hand can exist in infinitely many states.

Consider a basic qubit in the abstract: $|\psi\rangle$. Since this is a two dimensional system, we can write it as a vector $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where the $|0\rangle$ and $|1\rangle$ states form the Z orthonormal basis (also called the computational basis) as defined in Section 2.1. This state exists in a *superposition* of the state $|0\rangle$ and the state $|1\rangle$. Normalization requires that $\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1$. If $\alpha = 1$, then $|\psi\rangle = |0\rangle$; otherwise, if $\beta = 1$, then $|\psi\rangle = |1\rangle$. The α and β coefficients (which are complex numbers) are called the *probability amplitudes*. Thus, a qubit, like a c-bit, can exist in a “zero” or “one” state. However, α and β are any complex numbers subject to the before mentioned normalization constraint. There are actually infinitely many possibilities for the qubit to exist in. This property of superposition is the first major difference between a c-bit and a quantum bit (or general, quantum state).

2.3 MEASURING A QUANTUM STATE

The second major difference in qubits versus c-bits is how we extract information from them. With a c-bit, we simply read the bit, extract its state, and leave it undisturbed. Qubits, however, are different - reading a qubit, called *measuring* it, is a process that produces a potential random output and potentially destroys the original state.

Information may be extracted from a quantum state through measurements. These measurements lead to a discrete classical output, the result of which is potentially random and also potentially destructive to the state.

Consider the following state:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle). \quad (2.30)$$

The above is actually an important state which we'll refer to often - it's often called the “Hadamard plus state” or the “X basis plus state” (we'll use both interchangeably, or, simply $|+\rangle$). This state is in a superposition of both the $|0\rangle$ state and the $|1\rangle$ state. In a way, it is both the zero and one state simultaneously, though it is not quite as simple as that, especially when it comes time to *extract* information from a qubit, a process known as *measuring*.

A quantum state $|\psi\rangle$ remains in a “closed box” perhaps evolving its state according to some process we describe later in Section 2.4, perhaps in a superposition of multiple states, but, ultimately, remaining sealed away from the environment. Eventually, with any quantum algorithm or quantum cryptographic protocol, one must “look” at the state to extract information from it. This pro-

cess, called measuring, is the quantum analogue of reading a c-bit. However, while a quantum state may exist in a multitude of superposition states (e.g., the $|+\rangle$ state above, existing in a way as both zero and one), measurements cannot necessarily extract all information from a quantum state (e.g., cannot extract exact information on α and β for an arbitrary state). Instead, a quantum state, through measurement, is converted into classical data with discrete values (again, everything is finite dimensional in this text). This process is random and also leads to *state collapse*.

Consider a c-bit $x \in \{0, 1\}$. If this is stored in some classical memory system, one may read the bit at any time to get a deterministic output; furthermore the act of looking at the bit should not affect its state - it remains x . Consider, now, a quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. “Looking” at this state (i.e., measuring it) in the $|0\rangle, |1\rangle$ basis (we will see later, that we can look at a state in different bases - but for now, let’s stick to the standard computational basis) causes the measurement device to output *not* $\alpha|0\rangle + \beta|1\rangle$ (i.e., it will not output exact information on the complex numbers α and β); instead the measurement will output either “ $|0\rangle$ ” or “ $|1\rangle$ ” - a discrete, essentially classical outcome. This is a probabilistic process with the measurement device outputting (or *observing*) $|0\rangle$ with probability $|\alpha|^2$; otherwise, with probability $|\beta|^2$, the measurement device outputs $|1\rangle$.

This is the reason behind the name *probability amplitude*: the coefficient of a basis vector determines the probability of observing that particular basis element. The argument extends to multiple dimensions. For instance, if we have a state $\alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle$, then the probability of observing a “2” is $|\gamma|^2$. Note that normalization requires the sum, squared, of all probability amplitudes to be one, and thus this forms a valid probability distribution.

Exercise 2.4. Consider the following, three questions:

1. Consider the $|+\rangle$ state. What is the probability of observing $|1\rangle$ after a measurement?
2. Next, consider the following state:

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \quad (2.31)$$

What is the probability of observing $|1\rangle$ after measuring that state?

3. Finally, consider the state $|1\rangle$. What is the probability of observing $|1\rangle$ after measurement?

Beyond the fact that measurements are now random (unless the state happens to be one of the basis states, though we’ll comment more on this later), they also are destructive. The act of looking at a quantum state potentially disturbs it. In particular, if after making an observation, if your measurement observes outcome $|i\rangle$, then the state actually becomes $|i\rangle$. That is, the state *collapses* to the observed state vector; this is known as *state collapse*. Measurements,

therefore are: (1) random, (2) destructive and (3) non reversible (once measured, you cannot go back, reverse the measurement, and return to the state prior to measurement).

For example, let's say we start with the state $|+\rangle$. If we measure it, we will observe $|0\rangle$ or $|1\rangle$, both with equal probability. However, measurement is a “one-shot” deal. Once the qubit (originally in state $|+\rangle$) is measured, it collapses to the observed state. Thus, if our measurement produced an observation of “1”, the state now really is $|1\rangle$, the original $|+\rangle$ is “destroyed” and the state becomes $|1\rangle$. Any subsequent measurements will produce outcome “1” with probability one.

Exercise 2.5. Imagine you are given a qubit $|\psi\rangle$ but not told anything about it (i.e., you do not know anything about its probability amplitudes). If you perform a measurement on the state and the outcome is “0” what does this tell you about the state? Can you learn the original probability amplitudes? What if you are given as many copies of $|\psi\rangle$ as you want - what can be said then? Note that you cannot copy an unknown qubit yourself - we'll discuss this later though, in Section 2.5.4.

2.3.1 Basis Measurements

So far, quantum states and measurements, seem to provide a highly frustrating scenario for quantum information scientists. On the one hand, there are infinitely many states a single qubit can be in. On the other, to actually get information from a state, a measurement must be performed which creates an inherently classical outcome and, furthermore, causes the state to collapse to an, essentially, classical system (zero or one). Is there any real advantage to quantum states and superposition? The answer is, actually, yes though it will take a little more groundwork before it can be truly seen and appreciated. However, harnessing the power of quantum information, within the rules provided to us (e.g., within the realm of infinite state possibilities, but highly restricted measurement rules) is the ultimate challenge in any quantum related problem.

One important fact to point out is that measurements in the “zero/one” basis (the computational basis) are not the only form of measurement available to you. Indeed, one may measure in *any* basis of your state space! This opens up interesting possibilities as, now, there are infinitely many types of measurements one could perform on a qubit.

Consider a qubit, and let $\mathcal{B} = \{|b_0\rangle, |b_1\rangle\}$ be an orthonormal basis for the two dimensional qubit state space (namely, $\mathbb{C}^2$). It turns out, one can, at least in theory, take a qubit state $|\psi\rangle$, and measure in the $\mathcal{B}$ basis. (We say *in theory* since, in practice, actually engineering such a measurement may not always be as “easy” as describing it mathematically!)

How does this measurement work? As an example, let $|\psi\rangle$ be a qubit; since $\mathcal{B}$ is a basis, we may write the state as a linear combination of vectors in this

basis, namely:

$$|\psi\rangle = x |b_0\rangle + y |b_1\rangle. \quad (2.32)$$

for some $x, y \in \mathbb{C}$ with $|x|^2 + |y|^2 = 1$. Then, measuring in the $\mathcal{B}$ basis produces a discrete outcome: the measurement device observes either “ b_0 ” or “ b_1 .” Furthermore, the result is random, with the measurement observing $|b_0\rangle$ with probability $|x|^2$; otherwise the measurement device will observe $|b_1\rangle$, and that will occur with probability $|y|^2$. We still have state collapse: on observing “ b_i ” (for either $i = 0, 1$), the state, which was originally a superposition of these two basis vectors, will collapse to (i.e., become) the observed vector $|b_i\rangle$.

Of course, the above measurement rule of squaring the corresponding coefficient, only works when the input state is written with respect to the measurement basis. However, of course, this can always be done. Since $\mathcal{B}$ is a basis, given any state $|\psi\rangle$, one can always write that state vector as a linear combination (superposition) of basis states in $\mathcal{B}$. As an example, let $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$, and consider the *Hadamard basis*:

Definition 2.3.1. The *Hadamard basis* also known as the *X basis*, consists of two vectors $|+\rangle$ and $|-\rangle$ defined to be:

$$\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \\ |-\rangle &= \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \end{aligned}$$

Now, let’s take our state $|\psi\rangle$ and measure in the Hadamard basis. To do so, we must change basis and write $|\psi\rangle = x |+\rangle + y |-\rangle$. To do this, we will use the following identities:

$$\begin{aligned} |0\rangle &= \frac{1}{\sqrt{2}} |+\rangle + \frac{1}{\sqrt{2}} |-\rangle \\ |1\rangle &= \frac{1}{\sqrt{2}} |+\rangle - \frac{1}{\sqrt{2}} |-\rangle \end{aligned} \quad (2.33)$$

One can derive the above, by taking the definition of $|+\rangle$ and $|-\rangle$ and “solving for” $|0\rangle$ and $|1\rangle$. Using the above identities, we find:

$$\begin{aligned} |\psi\rangle &= \alpha |0\rangle + \beta |1\rangle = \alpha \left(\frac{1}{\sqrt{2}} |+\rangle + \frac{1}{\sqrt{2}} |-\rangle \right) + \beta \left(\frac{1}{\sqrt{2}} |+\rangle - \frac{1}{\sqrt{2}} |-\rangle \right) \\ &= \frac{\alpha + \beta}{\sqrt{2}} |+\rangle + \frac{\alpha - \beta}{\sqrt{2}} |-\rangle. \end{aligned}$$

From this, we see that our “ x ” is $(\alpha + \beta)/\sqrt{2}$ and our y is $(\alpha - \beta)/\sqrt{2}$. And, so, the probability of observing $|+\rangle$, given $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ is:

$$Pr(“+”) = \left| \frac{\alpha + \beta}{\sqrt{2}} \right|^2 = \frac{|\alpha|^2 + |\beta|^2 + \alpha\beta^* + \alpha^*\beta}{2} = \frac{1}{2} + Re(\alpha\beta), \quad (2.34)$$

where, above, we used the fact that $|\alpha|^2 + |\beta|^2 = 1$ and that, for any complex number z , it holds that $z + z^* = 2\text{Re}(z)$. We also used the fact that, for any two complex numbers z, w it holds that $(zw^*)^* = z^*w$.

Exercise 2.6. Let $p \in [0, 1]$ be a real number and consider the following state:

$$|\psi\rangle = \sqrt{p}|0\rangle + \sqrt{1-p}|1\rangle \quad (2.35)$$

Answer the following questions:

1. What is the probability of observing $|0\rangle$ for a given p ?
2. What is the probability of observing $|-\rangle$ for a given p ?
3. Graph the probability distribution of observing $|0\rangle$ with the x -axes of your plot being the value of p .
4. Same as above, but now graph the probability distribution of observing $|-\rangle$.

Thus, to measure $|\psi\rangle$ in basis $\mathcal{B}$, one must write out $|\psi\rangle$ as a superposition (linear combination) of basis vectors in $\mathcal{B}$. Once done, the probability of observing any particular basis element in $\mathcal{B}$ involves plucking out the corresponding coefficient of that vector (the probability amplitude) and take its norm squared.

Changing basis, though always possible, is not the only way to determine the probability distribution. Instead, one may look at the inner product. Let $\mathcal{B} = \{|b_0\rangle, \dots, |b_{d-1}\rangle\}$ be the orthonormal basis we want to measure in and let $|\psi\rangle$ be the d -dimensional state we want to measure. Then, if we denote by $p(b_i)$ to be the probability of observing “ b_i ” given this input state, it holds that:

$$p(b_i) = |\langle b_i|\psi\rangle|^2 = |\langle\psi|b_i\rangle|^2. \quad (2.36)$$

Note the distribution depends, of course, on the input state $|\psi\rangle$. The above equation is important to remember as it provides a much more natural way to compute the probabilities of measurement outcomes. One need only compute the inner product of the basis vector and the state you’re measuring. The “larger” the inner product, the more likely you are of observing that particular basis state. This gives a nice intuitive geometric interpretation of measurements.

But why does Equation 2.36 agree with our earlier interpretation of measurements - namely that the probability of making an observation is based on the probability amplitude, or the scalar coefficient “attached” to the basis vector in its linear combination. Well, let’s take $|\psi\rangle$ and since $\mathcal{B}$ is a basis, we may write it as:

$$|\psi\rangle = \sum_{j=0}^{d-1} \alpha_j |b_j\rangle. \quad (2.37)$$

From our earlier discussion, we know the probability of observing “ b_i ” is $|\alpha_i|^2$. So, to prove that Equation 2.36 matches this interpretation, we need to show that $\alpha_i = \langle b_i|\psi\rangle$. However, note that:

$$\langle b_i|\psi\rangle = \langle b_i| \times \left(\sum_{j=0}^{d-1} \alpha_j |b_j\rangle \right) = \sum_j \alpha_j \langle b_i|b_j\rangle = \sum_j \alpha_j \delta_{i,j} = \alpha_i, \quad (2.38)$$

where, above, $\delta_{i,j} = 1$ if $i = j$ and is zero otherwise. Thus, the two interpretations of measurement calculations – one using inner product and one using the scalar coefficients – give exactly the same values and may be used interchangeably. In general, the inner product form will be easier in the long run to use as you may start to easily run across similar calculations.

As a concrete example, let's compute the probability of observing a $|-\rangle$ given that an X basis measurement is performed on the state $|1\rangle$. For this, we compute:

$$\langle 1|-\rangle = \frac{1}{\sqrt{2}}(\langle 1|0\rangle - \langle 1|1\rangle) = -\frac{1}{\sqrt{2}}. \quad (2.39)$$

Thus, the probability of making this observation is $|\langle 1|-\rangle|^2 = 1/2$. (*Don't forget to take the norm square at the end!*) As a second example, let's make an X basis measurement of the state $|\psi\rangle = \sqrt{\frac{3}{4}}|0\rangle + \sqrt{\frac{1}{4}}|1\rangle$. What's the probability of observing “+”? We could make a change of basis as we did before and determine the correct probability amplitude. Instead, let's just use the inner product rule:

$$\langle +|\psi\rangle = \sqrt{\frac{3}{4}}\langle +|0\rangle + \sqrt{\frac{1}{4}}\langle +|1\rangle = \sqrt{\frac{3}{4}} \times \frac{1}{\sqrt{2}} + \sqrt{\frac{1}{4}} \times \frac{1}{\sqrt{2}}. \quad (2.40)$$

Note that, above, we made use of the fact that $\langle +|0\rangle = 1/\sqrt{2}$ and $\langle +|1\rangle = 1/\sqrt{2}$. This can be memorized (as it's useful), or easily derived. Continuing:

$$\langle +|\psi\rangle = \sqrt{\frac{3}{8}} + \sqrt{\frac{1}{8}} \implies p(+)=|\langle +|\psi\rangle|^2 = \frac{1}{2} + \frac{\sqrt{3}}{4}. \quad (2.41)$$

Exercise 2.7. Let $|\phi\rangle = \frac{1}{2}|+\rangle + \sqrt{\frac{3}{4}}|-\rangle$. Compute using both the inner product method and the change of basis method, the following:

1. $p(0)$
2. $p(-)$

Exercise 2.8. Let $|\psi\rangle$ be some d -dimensional state and let $\mathcal{B} = \{|b_0\rangle, \dots, |b_{d-1}\rangle\}$ be any orthonormal basis. Show that $|\psi\rangle$ may be written in the following form:

$$|\psi\rangle = \sum_{j=0}^{d-1} \langle b_j|\psi\rangle |b_j\rangle. \quad (2.42)$$

Note that, as before, anytime a measurement in this basis is made, the state collapses to the observed basis vector.

2.3.2 Measurement Operators

As it turns out, even more complicated measurements are possible, along with additional methods to compute the probabilities without having to directly make

a change of basis. Let $|\psi\rangle$ be a quantum state living in some state space $\mathcal{H}$. Let $\{M_x\}_{x \in \chi}$ be operators acting on the same state space which are indexed by some set χ (which, for our purposes, is finite). We will soon see that the set χ consists of observable outcomes that the measurement device might produce. Then, we say these operators are *measurement operators* if:

$$\sum_{x \in \chi} M_x^* M_x = I, \quad (2.43)$$

where I is the identity operator on $\mathcal{H}$. Then, we may measure $|\psi\rangle$ using these measurement operators. We will observe some outcome $x \in \chi$ with probability:

$$p(x) = \langle \psi | M_x^* M_x | \psi \rangle. \quad (2.44)$$

The post measured state collapses to:

$$\frac{M_x |\psi\rangle}{\sqrt{p(x)}}. \quad (2.45)$$

Measurement operators, those satisfying Equation 2.43, are the most general form of measurement allowed to us, and they generalize the basis measurements we've been considering so far. For example, a Z basis measurement is defined by measurement operators $M_0 = |0\rangle\langle 0|$ and $M_1 = |1\rangle\langle 1|$. Note that these are valid measurement operators since:

$$M_0^* M_0 + M_1^* M_1 = |0\rangle\langle 0| |0\rangle\langle 0| + |1\rangle\langle 1| |1\rangle\langle 1| = |0\rangle\langle 0| + |1\rangle\langle 1| = I, \quad (2.46)$$

where for the last equality we took advantage of the completeness relation discussed in Section 2.1.3. Now, let $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$, our typical, friendly, qubit state. From our earlier discussion, we know that the probability of observing $|0\rangle$ will be $|\alpha|^2$. As it turns out, this agrees with the measurement operator formalism. In particular:

$$\begin{aligned} p(0) &= \underbrace{(\alpha^* \langle 0| + \beta^* \langle 1|)}_{\langle \psi |} \underbrace{|0\rangle\langle 0| |0\rangle\langle 0|}_{M_0^* M_0} \underbrace{(\alpha |0\rangle + \beta |1\rangle)}_{| \psi \rangle} \\ &= \alpha^* \langle 0|0\rangle \alpha \langle 0|0\rangle = \alpha^* \alpha = |\alpha|^2. \end{aligned} \quad (2.47)$$

Note that the beauty of Dirac notation allowed us to readily compute the above in a very natural way by thinking of the above as a product of row and column vectors, leading to a 1×1 scalar. To be truly formal, one must recall the definition of an operator of the form $|v\rangle\langle w|$, from Section 2.1.3.

In general, measurement operators allow us to perform more ambitious measurements, beyond projective basis measurements. Measurement operators may also have more outcome possibilities than the dimension of the system. For example, one may design measurement operators which attempt to distinguish

two non orthogonal states, or output an “inconclusive” outcome. We’ll return to this shortly after discussing POVMs.

POVMs are *Positive Operator-Valued Measures* and they follow from the measurement operator formalism, however they are only useful if one is interested in the probability of an outcome occurring and not the actual post measured state. (Perhaps *only useful* is a bit strong as one can construct measurement operators from a POVM and thus compute the post measured state.) A POVM is a collection of operators $\{E_x\}_{x \in \chi}$, indexed over some outcome set χ , such that:

1. Each E_x acts on the state space of the system being measured.
2. Each E_x is a Hermitian, positive semi-definite operator (i.e., $E_x \geq 0$; see Section 2.25).
3. Finally, $\sum_{x \in \chi} E_x = I$.

Given a POVM $\{E_x\}$, and a quantum state $|\psi\rangle$, the probability of observing some outcome x is found to be:

$$p(x) = \langle \psi | E_x | \psi \rangle. \quad (2.48)$$

That the POVM formalism follows from measurement operators is easily seen. Let $\{M_x\}$ be a collection of measurement operators. Then, define $E_x = M_x^* M_x$. This is a POVM since each $E_x \geq 0$ (see Exercise 2.10). Furthermore, since $\{M_x\}$ are valid measurement operators, it follows that $I = \sum_x M_x^* M_x = \sum_x E_x$. Finally, if one were to measure using $\{M_x\}$, by Equation 2.44, the probability of observing outcome x is easily seen to match Equation 2.48.

Exercise 2.9. Show that each E_x is positive semi-definite, where $E_x = M_x^* M_x$, as discussed in the preceding paragraph.

Note that POVMs only have a rule to compute the probability of an outcome, but they say nothing of the post measured state. Often, though not always, this is sufficient, especially when dealing with information theoretic problems as we only care about the classical outcome of a measurement and, once a measurement is performed, we no longer care about the quantum state itself. (This is, of course, not always the case.) However, given a POVM, one may construct a collection of measurement operators using the rule $M_x = \sqrt{E_x}$, where $\sqrt{E_x}$ is the matrix function discussed in Section 2.1.4. This construction, of course, is not unique and there may be other measurement operators that “match” the given POVM.

2.3.3 Example: State Discrimination

POVMs are very useful in many cases. For instance, let’s consider the following experiment: say a source prepares one of two quantum states, either it prepares a $|0\rangle$ or it prepares a $|+\rangle$. When given a qubit (one of the two), your goal is to determine which state was sent to you. Now, as it turns out, this task is impossible on it’s own, which we can easily prove using POVM formalism. Let’s assume we could design some kind of measurement which can always

distinguish between $|0\rangle$ and $|+\rangle$. Well, given the above discussion, we might as well only consider POVMs; thus, let's assume there exist a POVM $\{E_0, E_+\}$ satisfying the following:

1. $\langle 0|E_0|0\rangle = 1$. That is, if the state really is $|0\rangle$, then the measurement should produce an outcome “0” with probability one.
2. $\langle +|E_+|+\rangle = 1$. Same as above, but for the case when the input state is $|+\rangle$.
3. $\langle +|E_0|+\rangle = \langle 0|E_+|0\rangle = 0$. That is, the measurement should never report the wrong answer.

However, this leads to contradiction. Indeed, consider $0 = \langle 0|E_+|0\rangle\langle 0|E_+|0\rangle$. By Exercise 2.10, it must hold that $\langle 1|E_+|0\rangle = \langle 0|E_+|1\rangle = 0$. But, then, since we also need $1 = \langle +|E_+|+\rangle$, and since $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, we have:

$$\begin{aligned} 1 = \langle +|E_+|+\rangle &= \frac{1}{2}(\langle 0|E_+|0\rangle + \langle 1|E_+|1\rangle + \langle 0|E_+|1\rangle + \langle 1|E_+|0\rangle) \\ &= \frac{1}{2} \langle 1|E_+|1\rangle \leq \frac{1}{2}, \end{aligned}$$

where, for the last inequality, we used the fact that $\langle 1|E_+|1\rangle$ represents the probability that the measurement would produce outcome “+” given an input state of $|1\rangle$ which cannot be larger than one. This leads to the contradiction that $1 \leq \frac{1}{2}$ and, so, such a POVM cannot exist.

Exercise 2.10. Let E_+ be a Hermitian positive semi-definite operator. Show that if $\langle 0|E_+|0\rangle = 0$, then it must be that $\langle 1|E_+|0\rangle = 0$. Hint: Define the operator $\sqrt{E_+}$ and consider $\sqrt{E_+}|0\rangle$. Use the fact that $\langle v|E_+|w\rangle = \langle v|\sqrt{E_+}\sqrt{E_+}|w\rangle$, along with the fact that, for any vector $|x\rangle$, if $\langle x|x\rangle = 0$, then $|x\rangle$ is the zero vector.

Thus, this POVM cannot exist. *However* if we change the rules of the game to include a third possible measurement outcome, namely an “I don’t know” outcome, then suddenly the task becomes possible. Now, we have a POVM denoted by operators $\{E_0, E_+, E_?\}$, with the following constraints needed:

1. $\langle 0|E_0|0\rangle > 0$
2. $\langle +|E_+|+\rangle > 0$
3. $\langle +|E_0|+\rangle = \langle 0|E_+|0\rangle = 0$.

Note that the difference is that, now, the operator is never wrong in that it will never report an incorrect state; however it no longer has to always report a conclusive outcome. Instead, it is allowed an additional inconclusive, or “I don’t know” result, namely $E_?$. We can define this POVM in the following way:

$$E_0 = p |-\rangle \langle -| \quad (2.49)$$

$$E_+ = p |1\rangle \langle 1| \quad (2.50)$$

$$E_? = I - E_0 - E_+. \quad (2.51)$$

Note that we define E_0 by having it “look” for a state orthogonal to $|+\rangle$ (which will never be observed if the state actually is $|+\rangle$). Similar for E_+ . These operators

“look” for states orthogonal to the other one. It is clear that this satisfies our requirements so long as $p > 0$. Defining $E_?$ as we did is a nice “trick” to make them sum to identity (a requirement for POVMs). The only thing we need to prove, then, is that they are all positive operators; namely that for some p , the operators above are positive. Clearly if $p = 0$ this holds trivially, but then our measurement always outputs “I don’t know.” Can we do better?

Let’s consider E_0 first. Let $|\psi\rangle$ be any vector (not even necessarily normalized). Then: $\langle\psi|E_0|\psi\rangle = p \langle\psi|-\rangle \langle-|\psi\rangle = p |\langle\psi|-\rangle|^2 \geq 0$. Similarly for E_+ . Thus, for any $p > 0$, E_0 and E_+ are both positive operators; the trouble case is the $E_?$ operators since, if p is too large, this becomes a negative operator. However:

$$\begin{aligned} \langle\psi|E_?|\psi\rangle &= \langle\psi|(I - E_0 - E_+)|\psi\rangle = \langle\psi|\psi\rangle - \langle\psi|E_0|\psi\rangle - \langle\psi|E_+|\psi\rangle \\ &= \langle\psi|\psi\rangle - p |\langle\psi|-\rangle|^2 - p |\langle\psi|1\rangle|^2. \end{aligned}$$

By the Cauchy-Schwarz inequality (see Equation 2.16), it holds that $|\langle\psi|-\rangle|^2 \leq \langle\psi|\psi\rangle \langle-|- \rangle = \langle\psi|\psi\rangle$. Thus:

$$\langle\psi|E_?|\psi\rangle \geq \langle\psi|\psi\rangle - 2p \langle\psi|\psi\rangle = (1 - 2p) \langle\psi|\psi\rangle. \quad (2.52)$$

The right hand side of the above expression is non-negative whenever $1 - 2p \geq 0$ or whenever $p \leq \frac{1}{2}$. Thus, we can conclude a POVM exists which satisfies our distinguishing requirements, so long as we set p between zero and $1/2$. The higher the value of p , the “better” the POVM is at distinguishing. Note that alternative constructions exist; this particular construction is, in some sense, the easiest one to design.

POVMs are powerful constructs and very useful for proving (1) measurements do not exist which satisfy some requirements or (2) constructing a “convoluted” measurement.

Exercise 2.11. Repeat the above state discrimination example for two arbitrary states $|\psi\rangle$ and $|\phi\rangle$ where $0 < |\langle\psi|\phi\rangle| < 1$.

2.4 STATE EVOLUTION

The next topic we cover is *state evolution* which specifies how a quantum state may change with time, without a (destructive) measurement being performed. This will dictate what processes we can apply to a quantum state, besides directly observing it through a measurement. State evolution is a non-destructive and, in fact, reversible process.

Given a quantum state $|\psi\rangle$, then, the state may change in time (or *evolve*) to a state $|\phi\rangle$ where $|\phi\rangle = U|\psi\rangle$. Here, U is a unitary operator (see Definition 2.1.2) and, recall, the notation $U|\psi\rangle$ means applying linear operator U to the state vector $|\psi\rangle$ (or multiplying matrix U with vector $|\psi\rangle$ if U is represented in matrix form). Said differently, one may change a quantum state $|\psi\rangle$ to a new

state $|\phi\rangle$ only if there exists a unitary operator U such that $|\phi\rangle = U|\psi\rangle$. Of course, the transformation may be difficult to implement practically - from a quantum cryptographic perspective, this may matter for Alice and Bob, but may not matter for the adversary, Eve, depending on the security model.

Before describing some basic unitary operators we will use throughout this book, let's first discuss how to describe a unitary operator. There are three main methods and to demonstrate each, and to show the difference with each, we'll consider a particular unitary operator known as the *Hadamard Operator* denoted H . The first method is the matrix representation method. Here we will assume the standard basis for the Z basis, namely $|0\rangle = (1, 0)^T$ and $|1\rangle = (0, 1)^T$. Then, the matrix representation of a unitary operator is simply, well, a matrix - in the qubit case it is a 2×2 matrix. The Hadamard is defined in this form as:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (2.53)$$

The *action* of the operator, then, on a state $|\psi\rangle$ is simply the matrix multiplication of the operator with the vector, namely the matrix product of $U|\psi\rangle$. For the Hadamard operator, we see the following identities:

$$\begin{aligned} H|0\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle \\ H|1\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle \end{aligned}$$

That is, H maps $|0\rangle$ to the $|+\rangle$ state while mapping $|1\rangle$ to the $|-\rangle$ state.

Exercise 2.12. Show that $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$. Note that not all unitary operators are their own inverse, however every unitary operator is invertible.

This is one way to represent a unitary operator. A second way involves simply describing the operator's action on basis states. Note that linear operators are completely defined based on their action on basis states. Thus, this gives a clean way to describe an operator, using only Dirac notation, without the need for vector notation and matrices (which, as the systems increase in size, can lead to unbearably large matrices!). For the Hadamard, we may equivalently define it in the following way:

$$\begin{aligned} H|0\rangle &= |+\rangle \\ H|1\rangle &= |-\rangle \end{aligned} \quad (2.54)$$

And that's it! In general, we will refrain from using matrix and vector notation and use only Dirac notation when possible and, so, will prefer this method of describing operators over the matrix representation method.

Exercise 2.13. Show that $H|-\rangle = |1\rangle$ using Equation 2.54 and 2.54, and not the matrix representation. To illustrate, we will show how $H|+\rangle = |0\rangle$ can be shown:

$$H|+\rangle = H\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \quad (2.55)$$

$$= \frac{1}{\sqrt{2}}H|0\rangle + \frac{1}{\sqrt{2}}H|1\rangle \quad (2.56)$$

$$= \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle = |0\rangle, \quad (2.57)$$

where the last equality follows from Equation 2.33.

Finally, the third representation method is the outer-product representation method. Here, we write U as a sum of simple operators of the form $|v\rangle\langle w|$, for different vectors $|v\rangle$ and $|w\rangle$. Recalling the action of an operator of the form $|v\rangle\langle w|$ on some vector $|x\rangle$ from Section 2.21, this allows us to write U as sums of such terms. To illustrate, consider, again, the Hadamard, which we now write as:

$$H = |+\rangle\langle 0| + |-\rangle\langle 1|. \quad (2.58)$$

Note, in particular, we wrote it as a sum of [“output”] [“input”] where we want a particular “input” (e.g., $|0\rangle$) to map to a particular “output” (e.g., $|+\rangle$). This way of thinking will always work if the “input” and “output” are orthonormal vectors.

Exercise 2.14. Let U be a unitary operator. Show that it can always be written in outer product form.

What are some example qubit (i.e., 2×2) unitary operators that are commonly used? The following list demonstrates a few that we’ll use throughout (besides the Hadamard operator which we’ve discussed in depth so far):

- *Pauli Z Operator:*

$$Z = \sigma_Z = |0\rangle\langle 0| - |1\rangle\langle 1| \quad (2.59)$$

- *Pauli X Operator:*

$$X = \sigma_X = |1\rangle\langle 0| + |0\rangle\langle 1| \quad (2.60)$$

- *Pauli Y Operator:*

$$Y = \sigma_Y = i|1\rangle\langle 0| - i|0\rangle\langle 1| \quad (2.61)$$

There are several others that we’ll use and we’ll introduce them as needed. Of course, we also need to consider unitary operators on two (or more) qubits. However, before describing that, we will need to show how to formally model multiple qubits. This is done using the tensor product, which we’ll define and discuss in Section 2.5.

2.4.1 Unitary Change in Basis

Let's say we're working in an n -dimensional space and let $\{|v_1\rangle, \dots, |v_n\rangle\}$ be an orthonormal basis. Let U be some unitary operator acting on this space. Then, abstractly, we may write the output of $U|v_i\rangle$ as simply being some vector we'll call $|w_i\rangle$. Namely, $|w_i\rangle = U|v_i\rangle$. What can we say about $|w_i\rangle$? As it turns out, the vectors $\{|w_1\rangle, \dots, |w_n\rangle\}$ also form an orthonormal basis.

Indeed, unitary operators *must preserve inner norm*. For any vectors $|x\rangle$ and $|y\rangle$, let $|v_x\rangle = U|x\rangle$ and $|v_y\rangle = U|y\rangle$. Then, it holds that:

$$\langle x|y\rangle = \langle x|I|y\rangle = \langle x|U^*U|y\rangle = \langle v_x|v_y\rangle. \quad (2.62)$$

If those vectors happen to be orthonormal, then:

$$\delta_{i,j} = \langle v_i|v_j\rangle = \langle v_i|I|v_j\rangle = \langle v_i|U^*U|v_j\rangle = \langle w_i|w_j\rangle, \quad (2.63)$$

where $\delta_{i,j}$ is the Kronecker delta function. The above follows since $U^*U = I$ by definition. From the above, it is not difficult to show that all the $|w_i\rangle$ are linearly independent also; thus they form an orthonormal basis.

Exercise 2.15. Show that if $\{|w_1\rangle, \dots, |w_n\rangle\}$ are orthonormal, then they are linearly independent.

Thus, unitary operators are simply changes in basis from one orthonormal basis to another. Indeed, given any two orthonormal bases, one may always find a unitary operator that maps basis vectors from the first to basis vectors from the second (hint: consider the outer-product formulation as in Equation 2.21).

This gives an alternative view to measurements. Consider a state $|\psi\rangle$ and we want to measure in some orthonormal basis $\{|w_1\rangle, \dots, |w_n\rangle\}$. Then, instead of looking at $|\langle w_i|\psi\rangle|^2$, we can instead apply a unitary operator U which takes $|w_i\rangle$ to the computational state $|i\rangle$. Namely apply $U = \sum_i |i\rangle\langle w_i|$ to $|\psi\rangle$ and then measure in the Z basis.

Why does this work? Well, since $\{|w_i\rangle\}$ is an orthonormal basis, we can write $|\psi\rangle = \sum_i \alpha_i |w_i\rangle$. Then, we know that the probability of observing basis vector $|w_i\rangle$ is simply $|\langle w_i|\psi\rangle|^2 = |\alpha_i|^2$. However, by applying the above operator U , we have:

$$U|\psi\rangle = \sum_i \alpha_i U|w_i\rangle = \sum_i \alpha_i |i\rangle. \quad (2.64)$$

Thus, now, the probability of observing $|i\rangle$ is, again, simply $|\alpha_i|^2$. This gives an alternative method for a basis measurement: instead of measuring in some orthonormal basis, one may first apply a unitary operator and then measure in the standard computational basis. The probability distribution will be identical. Note that the post-measured state will be a computational state in the latter case; however applying U^* will fix that.

Why is this important? From a computational standpoint, we want to argue about “how complicated” a quantum algorithm is. Normally a quantum

algorithm consists of applying several “small” unitary operators to a quantum state and then measuring it. However, this measurement may be complicated to describe! So, instead, if we force all algorithms to measure in the computational basis, we waive our hand over that issue - instead the complexity of the measurement will be absorbed into the initial circuitry by attempting to construct a change of basis unitary. This allows us to gauge the complexity of a quantum process, in terms only of the unitary circuitry and not the final measurement.

Exercise 2.16. Let $|\psi\rangle = \sqrt{\frac{3}{4}}|0\rangle - \sqrt{\frac{1}{4}}|1\rangle$. Note that the Hadamard operator may be described as $H = |+\rangle\langle 0| + |-\rangle\langle 1|$. Thus, it is a unitary change in basis from the Z basis to the Hadamard $+/-$ basis.

1. Using the inner-product rule of measurement, what is the probability of observing $|+\rangle$ and $|-\rangle$ if the above state $|\psi\rangle$ were measured in the Hadamard basis?
2. Now, apply H to $|\psi\rangle$. What is the probability of observing $|0\rangle$ and $|1\rangle$? Note that these pairs of numbers should match that which you computed above.

2.5 COMBINING SYSTEMS: TENSOR PRODUCT

Let's say we have a quantum state $|\psi\rangle$ and another quantum state $|\phi\rangle$. How do we model the joint system of two states together? This is done, mathematically, using tensor product and is denoted $|\psi\rangle \otimes |\phi\rangle$. The tensor space is a very abstract thing - for our sake we will not be too concerned with all the properties and the rigorous definition of what it means to tensor two vectors or two vector spaces together. Instead, we will rely on the basic terminology needed to analyze basic quantum algorithms and cryptographic protocols, introducing more advanced concepts as needed.

What does $|\psi\rangle \otimes |\phi\rangle$ “look like” mathematically? As it turns out, tensor product obeys many of the natural multiplicative rules you're already used to. If $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and $|\phi\rangle = x|0\rangle + y|1\rangle$, then we may think of $|\psi\rangle \otimes |\phi\rangle$ as the product of the two in the following way:

$$|\psi\rangle \otimes |\phi\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) = \alpha\gamma|0\rangle \otimes |0\rangle + \alpha\delta|0\rangle \otimes |1\rangle + \beta\gamma|1\rangle \otimes |0\rangle + \beta\delta|1\rangle \otimes |1\rangle. \quad (2.65)$$

Notice the similarity to, say, polynomial multiplication: if you had a polynomial $f(x, y) = \alpha x + \beta y$ and $g(x, y) = \gamma x + \delta y$, then the polynomial $f(x, y) \times g(x, y)$ is easily seen to be $\alpha\gamma x^2 + \alpha\delta xy + \beta\gamma yx + \beta\delta y^2$. Thinking of “ x ” as $|0\rangle$ and x^2 as $|0\rangle \otimes |0\rangle$, you can begin to see the relation. This may be a handy trick to remember the rules of tensor: for the most part, you can think about multiplying polynomials together. Though, keep in mind, we do not have commutativity, for instance $|0\rangle \otimes |1\rangle \neq |1\rangle \otimes |0\rangle$.

Now, notationally, Equation 2.65 is incredibly clunky. Tensor product is going to be a fairly common operation and so we don't want to have to write $\otimes$ continually (any more than we want to write $\times$ when doing standard multipli-

cation). So, we often drop the $\otimes$ and simply write $|\psi\rangle|\phi\rangle$ to mean $|\psi\rangle \otimes |\phi\rangle$. We may even go so far (and often do) as to write $|\psi, \phi\rangle$ to mean the same thing. Using this notation, we can write the above more succinctly as:

$$|\psi, \phi\rangle = \alpha\gamma|0, 0\rangle + \alpha\delta|0, 1\rangle + \beta\gamma|1, 0\rangle + \beta\delta|1, 1\rangle, \quad (2.66)$$

which is clearly preferred to the original form. Note that when dealing with multiple 0's and 1's inside a single ket (indicating multiple qubits together), we often may even forgo writing the comma; namely the above may be: $|\psi, \phi\rangle = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$. Note this only makes sense when we specifically have 0's and 1's; dropping the comma does not work well for, say, $|\psi\phi\rangle$ which we will never, deliberately, do.

Let's think, now, more abstractly, about what it would mean to have two qubits. A single qubit can be in a state $|0\rangle$, $|1\rangle$, or a superposition of both as discussed earlier. Then, it only makes sense that if we had *two* qubits, those two qubits could be in a state $|0, 0\rangle$, $|0, 1\rangle$, $|1, 0\rangle$, or $|1, 1\rangle$. However, as it turns out, these two qubits can also be in a superposition of all four states:

$$\alpha_{0,0}|0, 0\rangle + \alpha_{0,1}|0, 1\rangle + \alpha_{1,0}|1, 0\rangle + \alpha_{1,1}|1, 1\rangle. \quad (2.67)$$

The normalization rules apply as before, forcing the constraint $|\alpha_{0,0}|^2 + \dots + |\alpha_{1,1}|^2 = 1$.

Tensor product, as mentioned already, obeys a lot of the multiplication rules we're used to. In particular:

- $|\psi\rangle \otimes (|\phi\rangle + |\mu\rangle) = |\psi, \phi\rangle + |\psi, \mu\rangle$ for any $|\psi\rangle$, $|\phi\rangle$ and $|\mu\rangle$ (note that $|\phi\rangle$ and $|\mu\rangle$ have to live in the same underlying vector space. That is, tensor obeys distributive property.
- $\alpha(|\psi, \phi\rangle) = (\alpha|\psi\rangle) \otimes |\phi\rangle = |\psi\rangle \otimes (\alpha|\phi\rangle)$ for any scalar $\alpha \in \mathbb{C}$ and any two states $|\psi\rangle$ and $|\phi\rangle$. Namely scalars can be absorbed into any of the underlying vectors and the state is the same; equivalently, scalars may be “pulled out” of a tensor.

To demonstrate, consider $|0\rangle \otimes \frac{1}{\sqrt{2}}|0\rangle + |1\rangle \otimes \frac{1}{\sqrt{2}}|1\rangle$. We may “pull out” the scalar inside the inner system and take advantage of the distributive property to yield the equivalent state $\frac{1}{\sqrt{2}}|0, 0\rangle + \frac{1}{\sqrt{2}}|1, 1\rangle$.

If the reader likes Cartesian coordinate vectors, then there is a way to describe a tensor product of two vectors as a Cartesian coordinate vector itself. In particular, given:

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} \quad |\phi\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{pmatrix}, \quad (2.68)$$

namely two states, one of dimension n and the other of dimension m , then the tensor product $|\psi, \phi\rangle$ may be represented as a Cartesian vector using the

Kronecker Product. For this, we essentially multiply the right-most state “into” the left as follows:

$$|\psi, \phi\rangle = |\psi\rangle \otimes |\phi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} \otimes |\phi\rangle = \begin{pmatrix} \alpha_1 |\phi\rangle \\ \alpha_2 |\phi\rangle \\ \vdots \\ \alpha_n |\phi\rangle \end{pmatrix} = \begin{pmatrix} \alpha_1 \beta_1 \\ \alpha_1 \beta_2 \\ \vdots \\ \alpha_1 \beta_m \\ \alpha_2 \beta_1 \\ \alpha_2 \beta_2 \\ \vdots \\ \alpha_2 \beta_m \\ \vdots \\ \alpha_n \beta_1 \\ \alpha_n \beta_2 \\ \vdots \\ \alpha_n \beta_m \end{pmatrix} \quad (2.69)$$

where, above, each $\alpha_i |\phi\rangle$ represents an m -dimensional vector, namely the scalar product of α_i with $|\phi\rangle$. Note that the dimension of the new system is $n \cdot m$; that is the dimension of multiple quantum states goes up *multiplicatively*.

Note that tensor product, along with Kronecker product, can be applied multiple times which will be used to model three or more systems. The rules apply in the same way. Note that the dimension increases multiplicatively. Thus, if we have one qubit, the dimension of the state space is two; if we have two qubits, the dimension is $2 \times 2 = 4$; with three qubits we need a state space of $2 \times 2 \times 2 = 8$ and so on. The state space of n -qubits is 2^n -dimensional.

2.5.1 Linear Operators and Tensor Product

Tensor product has several nice properties when it comes to linear operators and their application to tensor states. First, let's consider the inner product. Let $|\psi\rangle = |a\rangle |b\rangle$ and $|\phi\rangle = |x\rangle |y\rangle$ be two vectors living in the same tensor space. The adjoint (or in our case, equivalently, the conjugate transpose since everything is finite dimensional) of $|\psi\rangle$ is simply: $\langle\psi| = \langle a| \langle b|$ (which may also be denoted $\langle a, b|$ or $\langle a| \otimes \langle b|$; all of these are equivalent). The interested reader may verify this identity by using the Kronecker product. The inner product, then, is $\langle\psi|\phi\rangle = (\langle a| \langle b|)(|x\rangle |y\rangle)$. Ultimately, we are applying a linear operator (the left $\langle a| \langle b|$) to the state $|x\rangle |y\rangle$. Whenever one has an operator $A \otimes B$ applying to a vector $|x\rangle |y\rangle$, you simply need to remember to apply the left-most operator to the left-most state and the right-most operator to the right most state. In general, if there are multiple tensors, say, $A_1 \otimes A_2 \otimes \cdots \otimes A_n$ applied to $|x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle$,

then one will apply A_i to $|x_i\rangle$. That is:

$$A_1 \otimes A_2 \otimes \cdots \otimes A_n |x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle = (A_1 |x_1\rangle) \otimes (A_2 |x_2\rangle) \otimes \cdots \otimes (A_n |x_n\rangle). \quad (2.70)$$

Of course, the map acts linearly on superpositions.

Returning to the inner product, therefore, we find:

$$\langle \psi | \phi \rangle = (\langle a | \langle b |) (|x\rangle |y\rangle) = \langle a | x \rangle \langle b | y \rangle. \quad (2.71)$$

Thus, the inner product of two simple-tensor vectors living in a tensor space is simply the product of the inner product of each individual subspace.

As a more concrete example, let's say $|\psi\rangle = |0, +\rangle$ and $|\phi\rangle = |- , 1\rangle$. Then:

$$\langle \psi | \phi \rangle = \langle 0, + | - , 1 \rangle = \langle 0 | - \rangle \langle + | 1 \rangle = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{2}. \quad (2.72)$$

Note, be careful when taking the conjugate transpose of a tensor: given the ket $|a, b\rangle$, then its adjoint operator is $\langle a, b|$; the order of the subspaces does not change: “a” is still the left-most, for instance.

What if we have states that are not simple tensors; namely what if we have more complicated states such as: $|\psi\rangle = \alpha |0, 1\rangle + \beta |+, -\rangle$ and $|\phi\rangle = \gamma |+, +\rangle + \delta |0, 0\rangle$? Then we may simply take advantage of the distributive property:

$$\begin{aligned} \langle \psi | \phi \rangle &= (\alpha \langle 0, 1 | + \beta \langle +, - |) (\gamma |+, +\rangle + \delta |0, 0\rangle) \\ &= \alpha \gamma \langle 0, 1 | +, + \rangle + \alpha \delta \langle 0, 1 | 0, 0 \rangle + \beta \gamma \langle +, - | +, + \rangle + \beta \delta \langle +, - | 0, 0 \rangle \end{aligned}$$

From the above, one may use the earlier rule for simple tensors to solve each individual case. We conclude:

$$\langle \psi | \phi \rangle = \alpha \gamma \frac{1}{2} + \alpha \delta \cdot 0 + \beta \gamma \cdot 0 + \beta \delta \frac{1}{2} = \frac{1}{2} (\alpha \gamma + \beta \delta). \quad (2.73)$$

The above rule of applying linear operators written in tensor form applies to any linear operator not just inner product. For example, if we have two unitary operators U_1 and U_2 , each of which act on a single qubit, then given a two qubit state $|\psi\rangle = \sum_{a,b} \alpha_{a,b} |a, b\rangle$, we can construct a unitary operator $U = U_1 \otimes U_2$ and its action will be as follows:

$$U |\psi\rangle = \sum_{a,b} \alpha_{a,b} U |a, b\rangle = \sum_{a,b} \alpha_{a,b} U_1 \otimes U_2 |a, b\rangle = \sum_{a,b} \alpha_{a,b} U_1 |a\rangle \otimes U_2 |b\rangle. \quad (2.74)$$

Namely, U_1 will apply to the first qubit while U_2 will apply to the second. Note that, given matrix representations of U_1 and U_2 , one may use the Kronecker product to construct a matrix representation of U ; however, we advise the reader not to bother and to try and think in terms of Dirac notation as matrix representations, though useful in some applications, quickly become tedious as the number of qubits advances.

2.5.2 Entanglement

Entanglement is one of the fundamental resources in quantum computation and communication, and is a fascinating topic on its own. At a, very rough, high level, two particles, one owned by Alice and one owned by Bob, are “entangled” if actions on one affect the state of the other, even if they are spatially separated. Even if they are separated by so great a distance so as to rule out any communication between the two particles. Formally, consider a two particle state $|\psi\rangle_{AB}$. We say it is *separable* if we can write $|\psi\rangle_{AB} = |a\rangle_A \otimes |b\rangle_B$; namely, if we can write the two-particle state as a simple tensor of two component systems. As an example, the state $|\psi\rangle_{AB} = \frac{1}{2}(|00\rangle_{AB} + |01\rangle_{AB} + |10\rangle_{AB} + |11\rangle_{AB})$ is separable, since we can write it as $|\psi\rangle_{AB} = |+\rangle_A \otimes |+\rangle_B = |+, +\rangle_{AB}$.

States that are separable are those which admit a very clear disconnection between each party; in particular, for a separable state, any action Alice performs on her system (e.g., a unitary operation on the A system, or a measurement of it) will clearly not affect the state of Bob’s particle. Indeed, just try tracing the execution of such an action: the unitary will be of the form $U_A \otimes I_B$ which will not affect Bob; similarly, her measurement operators will apply an identity to Bob’s particle and thus not affect his state; regardless, the state will remain $|a'\rangle_A \otimes |b\rangle_B$, where $|a'\rangle_A$ is the altered state after Alice interacts with her particle (i.e., $|a'\rangle = U_A |a\rangle$); Bob’s system remains the same.

Opposite to separable states are *entangled states* - i.e., an entangled system is one which is not separable. An important example of this is the so-called Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$. Why is this state entangled? And why is it the case that now, Alice’s operations will affect Bob? Let’s answer the first question first. We can show this is an entangled state by contradiction.

Assume, for contradiction, $|\Phi^+\rangle$ is not entangled; therefore it must be separable. If this is the case, we can write it as a simple tensor: $|\Phi^+\rangle = |a\rangle \otimes |b\rangle$ for some qubits $|a\rangle, |b\rangle$. The most general way we can write these two individual qubits is in the form $|a\rangle = \alpha |0\rangle + \beta |1\rangle$ and $|b\rangle = \gamma |0\rangle + \delta |1\rangle$. In this case, we have:

$$|a\rangle \otimes |b\rangle = \alpha\gamma |00\rangle + \alpha\delta |01\rangle + \beta\gamma |10\rangle + \beta\delta |11\rangle. \quad (2.75)$$

For the above to possibly be equal to $|\Phi^+\rangle$, it must be that either $\alpha = 0$ or $\delta = 0$ (to remove the $|01\rangle$ term which never shows up in $|\Phi^+\rangle$). But if $\alpha = 0$, then there is no $|00\rangle$ term! Therefore $\alpha \neq 0$ and it must be $\delta = 0$; but in that case, there is no $|11\rangle$ term. Thus, no solution exists and therefore $|\Phi^+\rangle$ is not separable, but entangled. Note that entangled states cannot be written as a single simple tensor; though, of course, they may be written as the sum of simple tensors!

Example 2.5.1. Consider the following:

1. $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$
2. $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$

Show that the first is separable but the second is entangled. To show a state is separable, write it as a simple tensor $|a\rangle \otimes |b\rangle$; to show it is entangled, prove by

contradiction that such a separation does not exist.

Now, let's assume Alice applies a unitary operation to her qubit; let's say the Hadamard operator H . Bob does nothing to his; thus, globally, we model this as the operator $H \otimes I_B$ and the state evolves to $H \otimes I_B |\Phi^+\rangle$. We claim that Alice performing her operation and Bob doing nothing, is identical to Bob applying H and Alice doing nothing! Clearly this is not the case with a separable system (please convince yourself of that, if you're not already, by tracing the execution of the operator to, say, $|++\rangle_{AB}$; you will get $|0+\rangle$ which is clearly not the same as $|+0\rangle$). Let's trace the execution of the operation on the entangled Bell state:

$$H \otimes I_B \left(\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle \right) = \frac{1}{\sqrt{2}} |+\rangle + \frac{1}{\sqrt{2}} |-\rangle \quad (2.76)$$

$$= \frac{1}{2} (|00\rangle + |10\rangle + |01\rangle - |11\rangle) = \frac{1}{\sqrt{2}} (|0+\rangle + |1-\rangle) \quad (2.77)$$

$$= I_A \otimes H \left(\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle \right) \quad (2.78)$$

Thus, unlike the separable case, actions on one particle affect the other. In this case, Alice applying a Hadamard to her qubit is equivalent to Bob applying it to his, even if he never performed any action on his system. This holds even if the two particles are spatially separated (potentially by billions of miles even in theory).

Exercise 2.17. Show that $H \otimes H |\Phi^+\rangle = |\Phi^+\rangle$. You can do this directly (work out the algebra) or you can use the above identity and the fact that $HH = I$. Also, and related, show that $|\Phi^+\rangle$ may be written in the X basis as $|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|++\rangle + |--\rangle)$.

2.5.3 Joint and Partial Measurements

Besides operators, measurements of one particle will affect the other if they are entangled. Let's look again at the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$. Without going into formal measurements (don't worry, we will!), let's just use our understanding of superposition states to try and determine the possible outcomes of different measurements. First, let's assume we have both particles in the same physical location; this would allow us to measure both systems simultaneously. If we were to measure in the two-qubit Z basis, spanned by $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, it is not difficult to see that we would observe $|00\rangle$ with probability $1/2$; otherwise, we would observe $|11\rangle$. This can be derived from the inner-product formulation of measurements, or using measurement operators, both of which were discussed in Section 2.3. Such a measurement on the entire state is typically called a *joint measurement*, and the rules discussed earlier apply (one must simply work over the new tensor state space).

Now, let's consider *partial measurements*, where only some, but not all, of the underlying systems are measured. For example, maybe two particles are spatially

separated, perhaps by a vast distance, thus prohibiting their joint measurement and forcing each particle to be measured individually. Or maybe one just wants to look at part of a system, but not the entire state.

Before going into the formalism, let's think about Alice making a measurement of the first system at a conceptual level. She will see either $|0\rangle$ or $|1\rangle$, each with probability $1/2$. Note since she is measuring only a single particle, she can only observe a two-dimensional outcome from such a basis measurement and the probability comes from squaring the appropriate coefficient in this specific case (in the general case it will be a bit more involved, but not significantly, as we'll soon discuss). Alternatively, if Bob had measured his particle he would observe the same possible outcomes with the same distribution.

However, and importantly, a measurement, even a partial one of spatially separated systems, causes state collapse of the entire, *global or joint* system! That is, in the first case, if Alice measures and observes $|0\rangle$, the entire system must collapse to a state that could agree with the observed reality. Since Alice saw $|0\rangle$, Bob's particle must collapse to $|0\rangle$ also, causing the joint state to become $|00\rangle$. Any subsequent measurement by Bob will always be $|0\rangle$. Similarly, if Alice observed $|1\rangle$, the entire state collapses to $|11\rangle$ and so Bob will subsequently always observe $|1\rangle$. Thus, even if the two particles are separated by vast distances, Alice and Bob will always get a correlated outcome, randomly either $|0\rangle$ or $|1\rangle$ (for both).

It must be noted that Alice and Bob are *not* communicating faster than light! Indeed, Alice gets a random outcome $|0\rangle$ or $|1\rangle$ same as Bob. While they know their random outcomes will be fully correlated, this does not constitute communication - namely, the distribution or outcomes cannot be "biased" based on some intended message to be sent.

While the above high-level discussion suffices for "simple" states (for example, repeat the above for the state $\frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$ - you should note that, now, Alice and Bob will always get opposite outcomes), it does not work well for more complex states. For instance, consider the *tripartite* state $\alpha|00e_0\rangle_{ABC} + \beta|10e_1\rangle_{ABC} + \gamma|+1e_2\rangle_{ABC}$, where the $|e_i\rangle$ states are sub normalized. Now, what is the probability of observing $|0\rangle$ if a measurement of the A system, in the Z basis, is performed? What is the probability of observing $|-\rangle$ if a measurement of the B system, in the X basis is performed? What about general measurements (e.g., measurement operators)? We'll actually answer the last question, which answers the first two (since basis measurements can be written as measurement operators).

Let's say Alice wishes to measure her system using some measurement $\{M_x\}_x$, where M_x are measurement operators (see Section 2.3.2), that act on Alice's state space (i.e., on Alice's system, not the entire joint state). Bob does not measure. In this case, we can define new measurement operators $\{\tilde{M}_x\}_x$, where $\tilde{M}_x = M_x \otimes I_B$. That these form a valid set of measurement operators is

straight-forward to show:

$$\begin{aligned}
 \sum_x \tilde{M}_x^* \tilde{M}_x &= \sum_x (M_x \otimes I_B)^* \otimes (M_x \otimes I_B) \\
 &= \sum_x (M_x^* \otimes I_B^*) (M_x \otimes I_B) \\
 &= \sum_x M_x^* M_x \otimes I_B = I_A \otimes I_B = I_{AB}.
 \end{aligned}$$

Above, we used the fact that $(A + B)^* = A^* + B^*$, along with the fact that $I_A \otimes I_B$ is the identity operator on the entire joint space, namely I_{AB} . Since we have a valid measurement operator to represent this partial measurement, all the rules from Section 2.3.2, describing the probability distribution of the measurement and any post-measured state, can be applied. Note that, on observing x , the post measured state becomes:

$$\frac{\tilde{M}_x |\psi\rangle_{AB}}{\sqrt{p(x)}} = M_x \otimes I_B |\psi\rangle_{AB} \sqrt{p(x)} \quad (2.79)$$

Sometimes, writing out explicit measurement operators is more work than needed. For example, in the simple case of $\alpha|0\rangle + \beta|1\rangle$, making a Z basis measurement using measurement operators is more work than simply looking at the probability amplitude and squaring it (though, of course, the end result is identical). When making a basis measurement of one system (i.e., a partial basis measurement), then there is a simpler approach. Let's assume we have a two-party state $|\psi\rangle_{AB}$ and we want to make a measurement of the A system in some orthonormal basis $\{|v_1\rangle, \dots, |v_d\rangle\}$. We could define measurement operators $M_i = |v_i\rangle\langle v_i|_A \otimes I_B$ and work out the algebra. Alternatively, if we write $|\psi\rangle_{AB}$ in the form:

$$|\psi\rangle_{AB} = \sum_{i=1}^d |v_i\rangle_A \otimes |b_i\rangle_B, \quad (2.80)$$

where the $|b_i\rangle_B$ states are arbitrary (potentially sub-normalized) vectors in the B state space, then it will follow that the probability of observing outcome $|v_i\rangle_A$ if a partial measurement is made on A , is simply $p(i) = \langle b_i | b_i \rangle$. The post measured state is $|v_i\rangle_A |b_i\rangle_B / \sqrt{p(i)}$. Note that, since $\{|v_i\rangle\}$ is an orthonormal basis of the A system, any state $|\psi\rangle_{AB}$ can be written in this form.

Exercise 2.18. Show that the “simple” form of a partial basis measurement, discussed above, will result in the correct distribution and post measured state, as dictated by the measurement rules using measurement operators.

2.5.4 No-Cloning Theorem

We'll close this section with an important result, namely the *no-cloning theorem*. Specifically, it can be shown that no unitary operator U exists which can

perfectly clone an arbitrary quantum state. (Actually, one can prove that no such *quantum operation* exists, and such an operation may consist of measurements, classical post-processing, and additional quantum ancillas.) Formally, let $|\psi\rangle$ be a quantum state, then to clone this state, we mean apply some quantum circuit (a unitary U) such that:

$$U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle. \quad (2.81)$$

where the input is the original state to clone, $|\psi\rangle$ and some “blank” ancilla initialized to the $|0\rangle$ state (where the clone will live, after the operation). The fact that this cannot be done, follows immediately from linearity (unitary evolution and measurement operators are all linear operators). Assume, for contradiction, that such an operation exists. Then, it must be that for two different states $|\psi\rangle$ and $|\phi\rangle$, it holds:

$$\begin{aligned} U |\psi\rangle |0\rangle &= |\psi\rangle |\psi\rangle \\ U |\phi\rangle |0\rangle &= |\phi\rangle |\phi\rangle \end{aligned}$$

However, let $|\tau\rangle = \alpha |\psi\rangle + \beta |\phi\rangle$ be some other quantum state, a superposition of both. If U is really a cloning operation, it should hold that $U |\tau\rangle |0\rangle = |\tau\rangle |\tau\rangle$. However, due to linearity, it actually holds that:

$$\begin{aligned} U |\tau\rangle |0\rangle &= U(\alpha |\psi\rangle + \beta |\phi\rangle) |0\rangle = \alpha U |\psi\rangle |0\rangle + \beta U |\phi\rangle |0\rangle \\ &= \alpha |\psi\rangle |\psi\rangle + \beta |\phi\rangle |\phi\rangle \end{aligned}$$

A simple calculation shows that

$$|\tau\rangle |\tau\rangle = \alpha^2 |\psi\rangle |\psi\rangle + \beta^2 |\phi\rangle |\phi\rangle + \alpha\beta |\psi\rangle |\phi\rangle + \beta\alpha |\phi\rangle |\psi\rangle. \quad (2.82)$$

Thus, $U |\tau\rangle |0\rangle$ can only be equal to $|\tau\rangle |\tau\rangle$ if $\alpha = 0$ or $\beta = 0$ and so U cannot clone any arbitrary state. Of course, one *can* clone basis states *if* one knows the basis the state is in. However, if the basis is unknown, then no cloner can exist which works for every state. This will be very important, later, when discussing security of various quantum cryptographic protocols.

2.6 QUANTUM INFORMATION THEORY

We now turn to the subject of quantum information theory, along with a discussion of some classical information theory to start off with. This is a broad, and truly fascinating field, and we can only scratch the surface, here. There are many, more detailed, texts on this subject, see, for instance, ?. Here, we only cover the tip of the iceberg, introducing topics that will be needed later.

Let’s first consider basic Shannon entropy. Imagine the following scenario: there is a source device that will transmit some message with some probability. In particular, let’s say the source will choose to send some message m_i , from a set

of possible messages $\{m_1, \dots, m_n\}$, choosing message m_i with some probability p_i . Now, imagine you are the receiver who knows the possible set of messages, and the probability distribution. You then receive a particular message m_i . How much information did you learn? This is the question (amongst others) Claude Shannon asked when developing his entropy function.

As an example, let's say there are two possible messages m_0 and m_1 and that the source sends m_0 with probability 0.99 (most likely) while sending message m_1 with probability .01 (very rarely). Now, let's say you, the receiver, receive message m_0 . Well, that really didn't tell you much - after all, you knew it was the most likely message to be sent. On the other hand, if you receive message m_1 , now that is very surprising and potentially gives a lot of information as that message was highly unlikely to arrive! This is the basic idea behind Shannon entropy: it will measure how much information, on average (or, alternatively, "how surprised you are") over the messages.

There are a few ways to define and build Shannon entropy - let's work with the *axiomatic approach* (see, for instance, ?). In particular, we want to consider, first, how much information we get from a message sent with probability p . First, we will note that, since the messages are labeled m_1 through m_n , the message itself is not so important as the index is. If we know the list of messages and send "5" we know we mean m_5 . Thus, the only important thing to consider, then, is the probability distribution and, so, the information function will be a function only of probability values p_i . (This may seem strange in practice when the message itself matters - however, if we know the messages and only care about which message we're receiving, then it is fairly clear why we can ignore the message labels and only consider the probability value of a message being sent.)

Let $I(p)$ be the information you attain on receipt of some message sent with probability p . Axiom (1): we will want information to be non-negative, thus $I(p) \geq 0$ for all $p \in [0, 1]$. Next, axiom (2), if a message shows up with probability one, then there is no information gained, thus $I(1) = 0$. Third (3), if two independent events occur, sending message m_i followed by m_j , choosing both independently (thus, their joint distribution is simply $p_i p_j$), the information gained is additive, namely $I(p_i p_j) = I(p_i) + I(p_j)$. Finally, (4) is that the information function should be continuous - namely, small changes to the probability value should not result in drastic changes in information. It can be shown that the above four axioms imply $I(p) = -\log_b p$, for some base b . The base will be the unit of measurement - here we will almost always have base two as we measure information in bits.

Now, consider the random variable M which chooses message i with probability p_i . The Shannon entropy of this random variable, denoted $H(M)$, is the expected information gain, namely:

$$H(M) = \sum_{i=1}^m p_i I(p) = - \sum_i p_i \log_2 p_i. \quad (2.83)$$

Above, we are simply assuming the base is two since, as mentioned, that is the case we care about in this book. Note that, sometimes, instead of writing $H(M)$, we may write $H(p_1, p_2, \dots, p_m)$. When $m = 2$ (i.e., there are only two outcomes or messages), then there is only one probability value to consider, since $p_2 = 1 - p_1$. In this case, we often write $h(p)$ to be the *binary Shannon entropy* - that is, the Shannon entropy of the two outcome experiment, with message one being sent with probability p and message two being sent with probability $1 - p$. Namely:

$$h(p) = -p \log p - (1 - p) \log(1 - p). \quad (2.84)$$

Note that $h(p) = h(1 - p)$ which may be easily verified. All logarithms in this section will be base two, unless otherwise specified.

While the above axiomatic construction is helpful, it is not immediately clear how this function can be useful for anything. As it turns out, it is highly important in a number of applications. Let's take, for instance, Shannon's *noiseless coding theorem* ???. Let's say a source is outputting a string of symbols (messages) with some probability distribution. Call this random variable M_1 . Then, after it chooses a symbol to send, it repeats, choosing another symbol with the same probability distribution, independently of the first. Call this (independent and identical) random variable M_2 . And so on producing an infinite series of symbols $M_1 M_2 \dots$ each one chosen independently and identically, of the others. Shannon asked, "what are the minimal number of bits that are needed to store the transmitted sequence so that it can be perfectly reconstructed later?" As it turns out the number of bits per symbol is exactly $H(M_i)$. As it turns out, this is the best one can do.

As a concrete example, say the source produces one bit (a zero or one) every round. If the bit chosen is zero with probability $1/2$ (otherwise it is one), then you will need $H(\frac{1}{2}, \frac{1}{2}) = 1$ bit per round to store the message. Not too surprising. However, now let's say that bit zero is chosen with probability $1/10$ (so, on average nine times out of ten, the source is sending a "1"). It should be intuitive that we can "compress" the received message in some way. Shannon's entropy can show exactly how much we can compress by - in this case, we only need $H(1/10, 9/10) = 0.469$ bits per round - a compression of almost 50%.

Moving to slightly more complex domains, consider now two random variables X and Y which may, or may not, be independent. We can talk about the *joint entropy*, denoted $H(XY)$, and it is defined in the natural way as:

$$H(XY) = - \sum_{x,y} p(x, y) \log p(x, y), \quad (2.85)$$

where $p(x, y)$ is the joint distribution of X taking the value x and Y taking the value y . If X and Y are independent, then:

$$H(XY) = H(X) + H(Y). \quad (2.86)$$

We can also talk about *conditional entropy*, namely the entropy of X conditioned on knowing the outcome of Y . This is denoted $H(X|Y)$ and is defined to

be:

$$H(X|Y) = H(XY) - H(Y). \quad (2.87)$$

You can think of it as the amount of information in both X and Y , subtracted by whatever information is in Y . It measures how much uncertainty we have on X , given that we know the outcome of Y . If X and Y are independent, then $H(X|Y) = H(X)$ (which you can prove in Exercise 2.19). If X and Y are completely correlated, then $H(X|Y) = 0$ since, given Y , you know exactly what X is. Conditional entropy will be very important later, when we discuss the security of QKD. In particular, we will be interested in how much uncertainty an adversary has on Alice or Bob's "key" - even if the adversary has some side information system that may be correlated with Alice or Bob.

Exercise 2.19. Let X and Y be independent random variables. Prove that $H(X|Y) = H(X)$. Similarly, show $H(Y|X) = H(Y)$. Why does this make intuitive sense?

Shannon entropy works well for classical random variables - that is, variables that take classical values based on classical probability distributions. But, now, how do we extend all of this to the quantum setting? For that, we will first introduce *von Neumann entropy*, followed by *quantum min entropy*. We'll first discuss von Neumann entropy as it is the most similar to Shannon entropy (and, in fact, is often called the "quantum analogue" of classical Shannon entropy). Later we'll move on to min entropy which, though more complicated to compute in general, is actually more useful from a cryptographic perspective. Before we can discuss any of these notions, however, we must first discuss *density operators*.

2.6.1 Density Operators

Up to this point, we have been working with *pure states* - i.e., systems that are completely self-contained and can be described, entirely, by their state vector. However, there are some instances where this is not the case. As a very basic example, tying well with the discussion above on Shannon entropy, consider a source device that creates quantum states probabilistically. In particular, the source will prepare state $|\psi_i\rangle$ with some probability p_i . Such a system cannot be described, naturally, using state vector formalism - especially when it comes time to describe exactly what the state of the system is when the receiver actually receives something. This is because the system that is sent is no longer "closed" - instead, it depends on the state of an external system (in this case the source device). This is where density operator formalism is useful - it can describe everything we have done so far (namely, pure states), but also much more. In general, a *quantum system* is a density operator which may, or may not, be pure. Note that, a state which is not pure is said to be *mixed* or a *mixed state*.

Density operator formalism allows us to describe scenarios such as the above (along with others - for instance, how do we describe an entangled system where

we only have access to one particle? Density operators can do that too!). Given the above example of a source producing some state $|\psi_i\rangle$ with probability p_i , the density operator describing that system is written:

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|, \quad (2.88)$$

where $|\psi_i\rangle \langle \psi_i|$ is the outer product of $|\psi_i\rangle$ (thus, it is a square matrix).

How do unitary operators work on a density operator? Well, going back to our source producing $|\psi_i\rangle$ with probability p_i (this is called an *ensemble of states*), let's imagine that the state produced gets sent to a "box" which implements some unitary U . Now, recall, the source box really is producing one of the $|\psi_i\rangle$ (it's not producing a superposition of different $|\psi_i\rangle$'s, nor is it producing multiple $|\psi_i\rangle$'s). Thus, at the output end of the "box" we will have state $U|\psi_i\rangle$ with probability p_i . Let $|\phi_i\rangle = U|\psi_i\rangle$. Thus, our density operator, describing the state after the unitary is performed, will be:

$$\sigma = \sum_i p_i |\phi_i\rangle \langle \phi_i| = \sum_i p_i U |\psi_i\rangle \langle \psi_i| U^* = U \rho U^*. \quad (2.89)$$

Above, we used the fact that $\langle \phi_i| = (U|\psi_i\rangle)^* = \langle \psi_i| U^*$. Thus, to apply a unitary operator to a density operator ρ , we multiply U on the left and U^* on the right.

What about measurements? Well, let $\{M_x\}$ be some measurement operators (we can introduce this using projective basis measurements, first, however we might as well go to the end goal and discuss general measurements, which will include the basis measurement case). Let $p(x|i)$ be the probability of observing outcome x given that the input state to the measurement box is $|\psi_i\rangle$, namely:

$$p(x|i) = \langle \psi_i | M_x^* M_x | \psi_i \rangle. \quad (2.90)$$

Now, it can be shown that $\text{tr}(M_x^* M_x |\psi_i\rangle \langle \psi_i|) = \langle \psi_i | M_x^* M_x | \psi_i \rangle$ using the *Gram Schmidt process* - see Exercise 2.20. Thus, taking into account the state is $|\psi_i\rangle$ with probability p_i , we may compute the total probability of observing x as simply:

$$p(x) = \sum_i p_i p(x|i) = \sum_i p_i \text{tr}(M_x^* M_x |\psi_i\rangle \langle \psi_i|) = \text{tr}(M_x^* M_x \rho), \quad (2.91)$$

In general, it can be shown, using the measurement rules from Section 2.3, along with basic probability theory, that the post-measured state, conditioned on observing " x " is the mixed state:

$$\rho_x = \frac{M_x \rho M_x^*}{\text{tr}(M_x^* M_x \rho)}. \quad (2.92)$$

Exercise 2.20. Let A be an operator and $|\psi\rangle$ be some vector. Show that:

$$\text{tr}(A |\psi\rangle \langle \psi|) = \langle \psi | A | \psi \rangle. \quad (2.93)$$

We introduced density operators using the notion of ensembles of quantum states: namely, using them to model random variables over quantum states. As it turns out, the resulting density operator is a positive Hermitian operator of unit trace (this is easy to verify). Hermitian operators were defined in Definition 2.1.3, while positive operators were defined in Definition 2.1.4.

On the other hand, every Hermitian operator is normal, thus diagonalizable; furthermore, it is easy to check that the eigenvalues in this case will be real (since it is Hermitian) and non-negative (since it is also positive semi-definite) and so may be written in the form $\rho = \sum_i \lambda_i |\psi_i\rangle \langle \psi_i|$, where $\lambda_i \geq 0$ and $|\psi_i\rangle$ are the eigenvalues and orthonormal eigenvectors, respectively. Thus, in this sense, every normal operator of unit trace can represent a quantum ensemble of some sort. Thus, we define any quantum state, or density operator, to be simply a Hermitian positive semi-definite operator of unit trace.

One important density operator that will occur often is the so-called *completely mixed state* and is simply the identity operator, acting on the state space, normalized by $1/d$ (where d is the dimension of the state space). Namely, I/d . For instance, with qubits, which are dimension two, the completely mixed state is $\frac{1}{2}I$. By the completeness relation, it holds that $I/2 = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$. Thus, this density operator represents the following ensemble: with equal, uniform, probability, the qubit is either $|0\rangle$ or $|1\rangle$ and, thus, is completely mixed. Note that, by the completeness relation, we could also write $I/2 = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -|$ and, so, it represents the system where the qubit is $|+\rangle$ with one-half probability, otherwise it is $|-\rangle$. We could also have chosen any other basis.

Now, the above brings up an interesting fact: Given an experiment (either a real experiment, or an idealized one), one can write the state as a density operator ρ . However, given ρ , there may be many other experiments that can be represented by the same exact operator! For instance, thinking of the previous paragraph, consider an experiment where a source device prepares a qubit in the state $|0\rangle$ or $|1\rangle$, choosing uniformly at random. The resulting density operator is $\rho = I/2$. Now, consider a completely different experiment where a source device prepares either $|+\rangle$ or $|-\rangle$, uniformly at random; the resulting density operator is $\sigma = I/2$. But, even though these are different experiments, the resulting density operator is the same. Namely, $\rho = \sigma$. How can this be and what does it mean? Keep in mind, for the first experiment, producing ρ , the qubit *really is* $|0\rangle$ or $|1\rangle$; whereas in the second experiment, producing σ , the qubit *really is* $|+\rangle$ or $|-\rangle$, two very different states. How can they produce the same density operator?

Well, let's look at it from a receiver's point of view. Let's say, a receiver device is going to perform some "quantum operations" on the received qubit (maybe some unitary operations, and a measurement, for instance). For the sake of illustration, let's just say a simple Z basis measurement is going to be performed on the incoming qubit. In the first case, with experiment one, producing $\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$, if the receiver measures in the Z basis, then the probability of getting a $|0\rangle$ is, of course, $1/2$ (because with $1/2$ probability the qubit really is a $|0\rangle$). Otherwise, the receiver will observe a $|1\rangle$ with equal

1/2 probability.

Now, let's consider the same exact measurement for the second experiment, producing $\sigma = \frac{1}{2} |+\rangle \langle +| + \frac{1}{2} |-\rangle \langle -|$. In this case, the qubit really is $|+\rangle$ with probability 1/2, otherwise it is $|-\rangle$. You may verify that the probability of the receiver observing $|0\rangle$ is, again, 1/2, same as the first experiment. The reader may readily verify that, even if the receiver applies a unitary operator, and then an arbitrary measurement, the statistics will be identical for both ρ and σ , even though they are two different experiments. Thus, even though these are two different physical experiments on the source side, the resulting outcome as far as the receiver is concerned (which is unaware of which experiment was performed), behaves identically for both systems. Density operators describe this behavior: when two operators are the same, even if they were produced by different systems, as far as the "owner" of the operator is concerned (e.g., the receiver in our case above), they will behave identically.

The above is very important from a cryptographic standpoint. Often, we will want to analyze a cryptographic protocol - doing so will allow us to write out a density operator ρ . However, sometimes, it is easier to consider an alternative protocol - this alternative protocol may not be the one that's actually run, however if it happens to produce the same exact state ρ (even though it did so using completely different operations than the first protocol), we may just as well analyze the second protocol. If two density operators are the same, even if they were produced through different means, there is no way to "distinguish" between them and the resulting output of the system will behave identically in both cases. This is perhaps one of the most subtle, and difficult to understand, part of density operators. It may take some time to really sink in, how important this concept is.

Another important, and useful, ability of density operators is their ability to model the classical and quantum output of a measurement. Let ρ_A be some quantum state and let's say we want to measure the A system using some measurement operators $\{M_x\}$ for $x \in \mathcal{X}$. Measurements, as we know, produce classical data (the measurement outcome) and quantum data (the post measured state). We can capture all of this in a single density operator which will model the classical data (the outcome of the measurement), the post measured quantum state, and finally the probability distribution of observing any particular outcome.

Let $\rho_A^{(x)}$ be the post measured state in the event one observes outcome x . Of course, from the discussion earlier (see Equations 2.91 and 2.92), we know $\rho_A^{(x)} = \frac{M_x \rho_A M_x^*}{p_x}$, where p_x is the probability of observing outcome x . Now, we will create a new space, denoted X , which will store the classical outcome. Let $\{|x\rangle : x \in \mathcal{X}\}$ be an orthonormal basis for this space (e.g., just use the standard basis if you like Cartesian vectors, however the exact basis doesn't matter so long as its orthonormal and fixed). The process will involve: measure the state resulting in a post-measured state, while saving the outcome in this new register. This can all be represented, compactly, as a single density operator which we'll

denote ρ_{XA} , as:

$$\rho_{XA} = \sum_{x \in \mathcal{X}} p_x |x\rangle \langle x|_X \otimes \rho_A^{(x)} = \sum_{x \in \mathcal{X}} |x\rangle \langle x|_X \otimes M_x \rho_A M_x^*. \quad (2.94)$$

A state in the above form, where one system is “classical” (i.e., written with respect to a single orthonormal basis and representing the storage of classical information), while the other is “quantum” (i.e., storing a quantum state), is often called a *classical-quantum state* or cq-state. They are used all throughout quantum cryptography, as, often, parties will be holding classical data (e.g., a secret key) while the adversary will be holding quantum side information (which may also include classical information depending on the adversary’s attack).

Note the notation of the density operators. The subscript XA represents a density operator acting on two subspaces: one for X and one for A . We will use this notation, throughout this text. For instance, a density operator σ_{ABE} represents a system (which may contain classical and quantum information), divided into three groups (or parties or subspaces, depending on the context). These groups are labeled A , B , and E . Such a system may be separable, or entangled; mixed, or pure. This notation will also extend to the following concept of *partial trace* where we may “trace out” parties, or systems, in a density operator.

2.6.1.1 Partial Trace

One of the powerful abilities of density operators is its ability to model quantum information that one party holds, given only part of a global system. For instance, let’s say Alice and Bob share an entangled particle pair in the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$. As a density operator, this state is represented by:

$$\rho_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}) \left(\frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}) \right)^* \quad (2.95)$$

$$= \frac{1}{2} (|00\rangle \langle 00| + |00\rangle \langle 11| + |11\rangle \langle 00| + |11\rangle \langle 11|) \quad (2.96)$$

where, above, we made use of the fact that the “bra” of a state vector is the conjugate transpose of that vector. Note, again, the subscript notation for the density operator, as commented on at the end of the last section.

This is a pure state (it is entangled, but taken together as an AB system, it is pure as it may be written as a state vector and, given the entire system description, there is no uncertainty in the state - it really is the Bell state). However, how do we mathematically represent the state that only Alice holds? Or only Bob holds? How do we “zoom in” on a particular sub-system of a larger state? This is one of the areas where density operators are highly useful as such systems can be represented through a *partial trace*.

Let’s say we want to zoom in on Alice (and ignore Bob). To write out the

density operator describing only Alice's system, we will write:

$$\rho_A = \text{tr}_B \rho_{AB}, \quad (2.97)$$

where tr_B means “trace over” system B (i.e., remove it from consideration but, importantly, don't remove its effect on the other systems). The partial trace is a linear operator, defined by the following action:

$$\text{tr}_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2| \text{tr}(|b_1\rangle\langle b_2|) = \langle b_2|b_1\rangle |a_1\rangle\langle a_2|. \quad (2.98)$$

Above, we used the fact that $\text{tr}(|x\rangle\langle y|) = \langle y|x\rangle$.

An easy way to perform a partial trace operation of some density operator ρ_{AB} is to write the system you're tracing out/over in an orthonormal basis. That is, if we're tracing out B , we can write ρ_{AB} in the form $\rho_{AB} = \sum_{b,b'} \rho_A^{(b,b')} \otimes |b\rangle\langle b'|$, where $|b\rangle$ is some orthonormal basis for the B state space. You can probably convince yourself that any arbitrary ρ_{AB} can be written in this form for any basis of your choice. Once that's done, using Equation 2.98 and the linearity property of partial trace, we have:

$$\begin{aligned} \text{tr}_B \rho_{AB} &= \text{tr}_B \left(\sum_{b,b'} \rho_A^{(b,b')} \otimes |b\rangle\langle b'| \right) \\ &= \sum_{b,b'} \text{tr}_B \left(\rho_A^{(b,b')} \otimes |b\rangle\langle b'| \right) \\ &= \sum_{b,b'} \rho_A^{(b,b')} \langle b'|b\rangle = \sum_b \rho_A^{(b,b)}. \end{aligned} \quad (2.99)$$

The last equality follows since $\langle b'|b\rangle = 0$ whenever $b \neq b'$ and, otherwise, $\langle b'|b\rangle = 1$.

Let's consider an example, namely the Bell state $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$, where $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Now, imagine the two particles are separated, so that Alice holds the A particle and Bob holds the B particle. If Alice were to measure in the Z basis, she would see $|0\rangle$ or $|1\rangle$, each with equal probability. If she were to measure in the Hadamard basis, she would also observe either basis vector with equal probability. One can confirm this, using the measurement rules discussed earlier (either for density operators, or for standard pure states).

But what does the partial trace look like (before Alice does anything)? Tracing over Bob reveals the following density operator:

$$\begin{aligned} \text{tr}_B(|\Phi^+\rangle\langle\Phi^+|) &= \text{tr}_B \left(\frac{1}{2} |00\rangle\langle 00|_{AB} + |00\rangle\langle 11|_{AB} + |11\rangle\langle 00|_{AB} + |11\rangle\langle 11|_{AB} \right) \\ &= \frac{1}{2} |0\rangle\langle 0|_A + \frac{1}{2} |1\rangle\langle 1|_A. \end{aligned}$$

For the above, note that:

$$|a, b\rangle\langle c, d| = |a\rangle\langle c|_A \otimes |b\rangle\langle d|_B. \quad (2.100)$$

This new density operator describes the system as far as Alice is concerned. Note it is a completely mixed state: measuring in the Z basis yields either $|0\rangle$ or $|1\rangle$ uniformly at random; similarly, measuring in the Hadamard basis also produces a uniform random output. This agrees with our intuition based on a partial measurement of the original pure state. Partial trace constructs a density operator which describes what can happen to a system if you only have that one system to work with and not the entire joint state.

Be careful: while this will correctly model Alice alone, one should keep in mind that Bob's particle has not "disappeared" from the universe. Instead, partial trace is a mathematical tool allowing us to zoom in on a particular party. There is, however, no way to go back to the original state if you don't keep track of the joint system. Partial trace will be very useful later when analyze quantum cryptographic protocols as we will often be interested in how systems behave according to a single party, even if that one party's system is entangled with a larger system. For instance, we will care about how an adversary's system behaves (and how much information it can contain), when it may be partially entangled with Alice and Bob's systems. More on this to come!

2.6.1.2 Trace Distance

One last topic we must cover concerning density operators is the *trace distance* which tells us how different (or, how similar) two quantum states are. Let ρ and σ be two density operators acting on the same space. Then, the trace distance between these two systems is denoted $\Delta(\rho, \sigma)$ and is defined to be:

$$\Delta(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\| = \frac{1}{2} \text{tr}|\rho - \sigma|, \quad (2.101)$$

where given some normal operator A , we write $|A|$ to mean $|A| = \sqrt{A^*A}$ (see Section 2.1.4 for details on functions such as this).

The above is a somewhat non-intuitive definition. Since our systems are finite dimensional, we can also write the trace distance as:

$$\Delta(\rho, \sigma) = \frac{1}{2} \sum_i |\lambda_i|, \quad (2.102)$$

where λ_i are the eigenvalues of $\rho - \sigma$. Note that if ρ and σ are exactly the same state, then $\rho - \sigma = 0$ (the zero operator) and so the eigenvalues are all zero and, thus, $\Delta(\rho, \sigma)$ would be zero. As the distance between ρ and σ increases, Δ will also increase.

Exercise 2.21. Show that Equations 2.101 and 2.102 are equivalent.

Trace distance has several important and useful properties. One is that it is invariant to unitary operations. In particular, let U be a unitary operator. Then:

$$\Delta(\rho, \sigma) = \Delta(U\rho U^*, U\sigma U^*). \quad (2.103)$$

Thus, applying a unitary operator to two quantum states cannot increase (or decrease) their distance from one another. Quantum operations in general (such as, for example, partial trace), cannot increase trace distance. For example, given ρ_{AB} and σ_{AB} it holds that:

$$\Delta(\rho_{AB}, \sigma_{AB}) \geq \Delta(\rho_A, \sigma_A). \quad (2.104)$$

Exercise 2.22. Let $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$ and $\sigma_{AB} = |\Phi^-\rangle\langle\Phi^-|$, where $|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle_{AB} \pm |11\rangle_{AB})$. Show that $\Delta(\rho_{AB}, \sigma_{AB}) = 1$ but $\Delta(\rho_A, \sigma_A) = 0$.

Another important property that holds is the *triangle inequality*. In particular for any density operators ρ, σ , and τ (acting on the same state space), it holds that:

$$\begin{aligned} \Delta(\rho, \sigma) &= \frac{1}{2} \|\rho - \sigma\| = \frac{1}{2} \|(\rho - \tau) + (\tau - \sigma)\| \\ &\leq \frac{1}{2} \|\rho - \tau\| + \frac{1}{2} \|\tau - \sigma\| = \Delta(\rho, \tau) + \Delta(\tau, \sigma). \end{aligned} \quad (2.105)$$

Trace distance has several other very important properties and uses. We'll see more as we continue in this text.

2.6.2 Quantum Entropy

Now that we have described density operators, we can resume our discussion on quantum information theory, turning first to von Neumann entropy. Let ρ_A be a quantum system (a density operator), acting on some Hilbert space $\mathcal{H}_A$ of dimension n . Then, we write $H(A)_\rho$ to be the von Neumann entropy of ρ_A (note the ρ subscript, to indicate this is von Neumann entropy, and not Shannon entropy; note that some authors write $S(\rho_A)$ instead). This is defined to be (see, for instance, ?):

$$H(A)_\rho = -\text{tr} \rho \log \rho. \quad (2.106)$$

Since ρ_A is a density operator, it is diagonalizable. Thus, we can write $\rho_A = \sum_i \lambda_i |v_i\rangle\langle v_i|$, where $|v_i\rangle$ form an orthonormal basis (the eigenvectors) and λ_i are real, non-negative, eigenvalues. Using matrix functions, discussed in Section 2.1.4, we have:

$$H(A)_\rho = -\text{tr} \rho \log \rho = -\sum_i \lambda_i \log_2 \lambda_i = H(\lambda_1, \dots, \lambda_n). \quad (2.107)$$

That is, von Neumann entropy is the Shannon entropy of the eigenvalues of the density operator. In a way, one may think of von Neumann entropy, as the uncertainty in a quantum state, assuming one makes an “optimal” measurement - i.e., a measurement in the eigenbasis of the operator.

Let's do a few examples. Consider the pure state $\rho_A = |0\rangle\langle 0|$. Intuitively, it seems like this state has no uncertainty to it (it is, after all, no different

mathematically from a classical “0” with probability one). And, indeed, we see that, since we can write $\rho_A = 1 |0\rangle\langle 0| + 0 |1\rangle\langle 1|$, we have $H(A)_\rho = H(1, 0) = 0$.

Now, what if $\rho_A = |+\rangle\langle +|$. Perhaps one might think that there is some uncertainty here, since this is a superposition state. However, we can write $\rho_A = 1 |+\rangle\langle +| + 0 |-\rangle\langle -|$ and, so, once again we find $H(A)_\rho = H(1, 0) = 0$. That is, there is no uncertainty in this state, despite the fact that it is a superposition state of the form $\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$. However, this is perhaps not too surprising once one thinks about it for some time: indeed, the basis label really makes no difference. If we have a pure state in the form $|0\rangle$, and we know it’s a $|0\rangle$, then there is no uncertainty. Similarly, if we have a pure state $|+\rangle$ and we know it is $|+\rangle$, then there is, again, no uncertainty. For the first, we can measure in the Z basis to retrieve the information correctly, while in the second we can measure in the X basis. Pure states, such as this, represent states on which we have complete knowledge - they really are in a particular state. It is only once we get to a mixed state where there can be some uncertainty.

Exercise 2.23. Let ρ_A be a pure state (i.e. $\rho_A = |\psi\rangle\langle \psi|$ for some vector $|\psi\rangle$ of arbitrary, but finite, dimension). Show that $H(A)_\rho = 0$. Similarly, show that, if $H(A)_\rho = 0$, then ρ_A must be pure. That is, prove the following:

$$H(A)_\rho = 0 \iff \rho_A \text{ is a pure state.} \quad (2.108)$$

As another example, now consider the mixed state $\rho_A = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1|$. In this case, we find $H(A)_\rho = H(\frac{1}{2}, \frac{1}{2}) = 1$. Again, not very surprising, since this density operator represents a “classical coin flip” - with one half probability the state is a “0” otherwise it is a “1”. But, as it turns out, the mixed state $\sigma_A = \frac{1}{2} |+\rangle\langle +| + \frac{1}{2} |-\rangle\langle -|$ also has von Neumann entropy equal to one.

Now, let’s consider $\rho_A = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |+\rangle\langle +|$. This state is not in a diagonal form (since $|0\rangle$ and $|+\rangle$ are not orthogonal). Thus, to compute the von Neumann entropy, we must diagonalize it. We can do so, numerically, and find that the eigenvalues of ρ_A are approximately 0.854 and 0.146 implying that the von Neumann entropy is 0.599.

Von Neumann entropy has several useful properties. In particular, it holds that $H(A)_\rho \geq 0$ and $H(A)_\rho \leq \log_2 \dim \rho_A$. The von Neumann entropy is maximal only on the purely mixed state I/d , where d is the dimension of the system. Von Neumann entropy is also concave. In particular, if $\rho_A = \sum_i p_i \rho_A^{(i)}$, then:

$$H(A)_\rho \geq \sum_i p_i H(A)_{\rho^{(i)}}, \quad (2.109)$$

where, by $H(A)_{\rho^{(i)}}$, we mean the entropy in the state $\rho_A^{(i)}$. Other properties will be brought up as needed in this book.

As with Shannon entropy, we can also talk about joint and conditional entropies. Let ρ_{AB} be a density operator acting on joint Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$. Then, we write $H(AB)_\rho$ to mean the entropy in the entire system. This will be

the Shannon entropy of the eigenvalues of ρ_{AB} . If we were to write $H(A)_\rho$, for a given ρ_{AB} , we would mean the entropy in the state $\rho_A = \text{tr}_B \rho_{AB}$. Finally, the conditional von Neumann entropy, denoted $H(A|B)_\rho$ is defined to be:

$$H(A|B)_\rho = H(AB)_\rho - H(B)_\rho, \quad (2.110)$$

where, similar to before, the last entropy is the entropy of $\rho_B = \text{tr}_A \rho_{AB}$.

In QKD security proofs (and other proofs of security), at least in the so-called *asymptotic setting*, we will need to find bounds on $H(A|E)$ for some state ρ_{AE} . Such a state is shared between Alice (the A register) and an adversary Eve (who holds the E register). Typically, such states are classical-quantum and may be written as:

$$p_0 |0\rangle \langle 0|_A \otimes \rho_E^{(0)} + p_1 |1\rangle \langle 1|_A \otimes \rho_E^{(1)}. \quad (2.111)$$

Above, $\rho_E^{(a)}$ is Eve's conditional state, namely her quantum state in the event Alice's bit register is $a \in \{0, 1\}$. Since these are quantum states, i.e., they are density operators, they are diagonalizable. Furthermore, they have real, non-negative, eigenvalues (see Section 2.6.1). In this case, we can write the above in the following form:

$$p_0 |0\rangle \langle 0|_A \otimes \left(\sum_{i=0}^N \lambda_i^0 |E_i^0\rangle \langle E_i^0|_E \right) + p_1 |1\rangle \langle 1|_A \otimes \left(\sum_{i=0}^N \lambda_i^1 |E_i^1\rangle \langle E_i^1|_E \right). \quad (2.112)$$

By “absorbing” all the scalars into Eve's state vectors, we may actually write the above more simply as:

$$\frac{1}{M} |0\rangle \langle 0|_A \otimes \left(\sum_{i=0}^N |F_i^0\rangle \langle F_i^0|_E \right) + \frac{1}{M} |1\rangle \langle 1|_A \otimes \left(\sum_{i=0}^N |F_i^1\rangle \langle F_i^1|_E \right) \quad (2.113)$$

where M is some normalization term (see Exercise 2.24 below).

The question, now, is what is $H(A|E)_\rho$? To compute this, we would need the eigenvectors of ρ_E , however the vectors $|F_i^j\rangle$ are not completely known. However, it can be shown that if some information is available on them, in particular if one knows the inner products $\langle F_i^j | F_i^j \rangle$ and also $\langle F_i^0 | F_i^1 \rangle$, then we can bound the von Neumann entropy. This is stated in Theorem 1 below which is easily proven (a proof may be found in ? which is where this particular wording of the theorem also comes from).

Theorem 1. (From ? for a proof): Given a state ρ_{AE} of the form in Equation 2.113, then it holds that:

$$H(A|E)_\rho \geq \sum_{i=1}^N \left(\frac{\langle F_i^0 | F_i^0 \rangle + \langle F_i^1 | F_i^1 \rangle}{M} \right) \times S_i, \quad (2.114)$$

where:

$$S_i = h \left(\frac{\langle F_i^0 | F_i^0 \rangle}{\langle F_i^0 | F_i^0 \rangle + \langle F_i^1 | F_i^1 \rangle} \right) - h(\lambda_i), \quad (2.115)$$

and, finally:

$$\lambda_i = \frac{1}{2} \left(1 + \frac{\sqrt{(\langle F_i^0 | F_i^0 \rangle - \langle F_i^1 | F_i^1 \rangle)^2 + 4Re^2 \langle F_i^0 | F_i^1 \rangle}}{\langle F_i^0 | F_i^0 \rangle + \langle F_i^1 | F_i^1 \rangle} \right). \quad (2.116)$$

This theorem will be helpful later, in Chapter 5, when we prove security of the BB84 QKD protocol against so-called “collective” attacks. Above, $h(x)$ is the binary Shannon entropy function (Equation 2.84).

Exercise 2.24. Show that Equation 2.112 may be written in the form depicted in Equation 2.113.

2.6.2.1 Quantum Min Entropy

Now, von Neumann entropy is highly useful when analyzing the security of a QKD protocol, in asymptotic scenarios. However, one other very important entropy will be *conditional quantum min entropy*, denoted $H_\infty(A|E)$. This is a “worst case” entropy measure that we will use a lot, later, when proving the security of various quantum cryptographic systems, in the *finite key* setting (the distinction between asymptotic scenario and finite key scenario will be made clear in Chapter 5).

Let ρ_{AE} be a quantum state, then the quantum min entropy is defined to be ?:

$$H_\infty(A|E)_\rho = \sup_{\sigma_E} \max \{ \lambda \in \mathbb{R} : 2^{-\lambda} I_A \otimes \sigma_E - \rho_{AE} \geq 0 \}, \quad (2.117)$$

where the above supremum is over all σ_E acting on the E state space (Hilbert space) and where, by $A - B \geq 0$, for operators A and B , we mean $A - B$ is *positive semi-definite*, defined in Definition 2.1.4. Compared to von Neumann entropy, this is, perhaps, not as intuitive a function. As it turns out, when ρ_{AE} is a cq-state (see Section 2.6.1) with $\rho_{AE} = \sum_x p_x |x\rangle \langle x|_A \otimes \rho_E^{(x)}$, then quantum min entropy has a surprisingly nice operational definition, as proven in ?. In this case, it turns out that the min entropy is related to the maximal guessing probability of Alice’s state, given Eve’s side information, namely:

$$H_\infty(A|E) = -\log p_g(A|E), \quad (2.118)$$

where $p_g(A|E)$ is the guessing probability:

$$p_g(A|E) = \max_{\{\mathcal{E}_x\}} \left(\sum_x p_x \text{tr} \mathcal{E}_x \rho_E^{(x)} \right). \quad (2.119)$$

The maximization, above, is overall all POVMs over Eve’s Hilbert space.

Sometimes, we will want to “discard” bad events which happen with a small probability. In this case, it is often useful to switch to *smooth min entropy*,

denoted $H_\infty^\epsilon(A|E)_\rho$, and defined to be ?:

$$H_\infty^\epsilon(A|E)_\rho = \sup_{\sigma_{AE}} H_\infty(A|E)_\sigma, \quad (2.120)$$

where the supremum is over all density operators σ_{AE} which are ϵ -close to ρ_{AE} , namely $\|\rho_{AE} - \sigma_{AE}\| \leq \epsilon$. This will be very useful, later, when analyzing QKD protocols as there may be certain events which, though occurring with small probability, would cause the min entropy to be very low, or zero, were they to occur. By switching to smooth min entropy, we may “discard” these events, though it must be done carefully, as we will see as this book progresses.

Let’s take a look at some examples. If $\rho_{AE} = \sum_x p_x |x\rangle \langle x|_A \otimes |x\rangle \langle x|_E$, then it can be shown that $H_\infty(A|E)_\rho = 0$ since Eve is fully correlated, Eve will know exactly what data Alice has. On the other hand, consider: $\sigma_{AE} = \sum_x p_x |x\rangle \langle x|_A \otimes |0\rangle \langle 0|_E$. In this case, Eve’s system is fully independent (separable) of Alice’s system, the best she can do is guess randomly at Alice’s value. The best strategy in this case is to assume Alice will get the value of x with the maximum p_x , and so $H_\infty(A|E)_\sigma = -\log \max_x p_x$.

Quantum min entropy is, in general, a difficult quantity to bound. However, there are some cases where it is easy, or “easier.” Besides entropic uncertainty, which we discuss below in Section 2.6.2.2, there are certain types of states where min entropy may be readily calculated or bounded. One interesting case is when we condition on a classical system. In particular, consider the state:

$$\rho_{AEC} = \sum_{c=0}^N p_c |c\rangle \langle c|_C \otimes \rho_{AE}^{(c)}. \quad (2.121)$$

This state has a classical register C where, if $C = c$, then the AE system is in the state $\rho_{AE}^{(c)}$. In this case, we have:

$$H_\infty(A|E)_\rho \geq H_\infty(A|EC)_\rho \geq \min_c H_\infty(A|E, C = c)_\rho \quad (2.122)$$

The first inequality actually applies to any state: namely, conditioning on another system can only decrease uncertainty. The second inequality applies for the specific state in question. Here, we write $H_\infty(A|E, C = c)_\rho$ to mean the min entropy in the conditional AE state when $C = c$, namely $H_\infty(A|E, C = c)_\rho = H_\infty(A|E)_{\rho^{(c)}}$. The above states that min entropy may be bounded by the “worst case” event. Von Neumann entropy has significantly better lower bounds in these cases; however, min entropy is, in a way, a “worst case” entropy measure (“worst case” in the terms of cryptography, where we would ideally want an adversary E to have a lot of uncertainty in general).

Another interesting case for min entropy is when we have a superposition of “small terms.” Namely, consider the following (pure) state where the A system consists of n qubits and the E system is arbitrary, and potentially entangled:

$$|\psi\rangle_{AE} = \sum_{i \in J} \alpha_i |i\rangle_A |E_i\rangle_E, \quad (2.123)$$

where $J \subset \{0, 1\}^n$. Now, let's assume a measurement of each qubit of A is made in some basis; let X be the random variable denoting this measurement and let the post measured system be ρ_{XE} (see Section 2.6.1). What is $H_\infty(X|E)_\rho$? This is not an easy question to answer. Now consider the following *mixed* state:

$$\sigma_{AE} = \sum_{i \in J} |\alpha_i|^2 |i\rangle \langle i| \otimes |E_i\rangle \langle E_i|. \quad (2.124)$$

Recall the difference between $|\psi\rangle_{AE}$ and σ_{AE} . The former is a superposition of states, while the latter is a mixed state (the A system really is $|i\rangle$ with probability $|\alpha_i|^2$). Now, let's say a measurement is made of the A system, using the same basis as before, of this mixed state, resulting in a new density operator σ_{XE} . In general, computing $H_\infty(X|E)_\sigma$ is straight-forward: one must consider the “worst case” $i \in J$ (using Equation 2.122) and follow the measurement on that. As it turns out, we can use this, along with $|J|$ to bound the entropy in the case the state was pure. In particular, it was shown in ? (based on a proof in ?), that:

$$H_\infty(X|E)_\rho \geq H_\infty(X|E)_\sigma - \log |J|. \quad (2.125)$$

If $|J|$ is small, the entropy of the pure-state case cannot be too different from the mixed state case. This property will be useful, later, when analyzing the security of various quantum cryptographic systems.

2.6.2.2 Entropic Uncertainty

In general, we will be dealing with systems of high dimensions, where Alice's system is of dimension 2^n and Eve's system is of arbitrary, but finite, dimension. In this case, computing the quantum min entropy is difficult, often more so than computing the von Neumann entropy (the latter of which admits several nice simplifications, and obeys some more structured properties than min entropy). However, there are several techniques that we can employ to at least bound min entropy, one of which is *entropic uncertainty*.

Before discussing entropic uncertainty for quantum min entropy, we'll discuss some basic uncertainty relations for von Neumann entropy (which are also very useful, especially in cryptographic settings). Let's assume we have a quantum state ρ_A and, furthermore, let's make it easy on ourselves and say ρ_A is a qubit (so, two dimensional). Let's also say we have two orthonormal bases $N = \{|n_0\rangle, |n_1\rangle\}$ and $M = \{|m_0\rangle, |m_1\rangle\}$. If we measure ρ_A in basis N we can consider the outcome to be a random variable and call the post measured (mixed) state ρ_N . Otherwise, if we measure in M we can call the post measured state ρ_M . Then, we can talk about the uncertainty contained in these two states (or, equivalently, the uncertainty in the final measurement outcome). As it turns out, the following expression always holds:

$$H(M) + H(N) \geq -\log c, \quad (2.126)$$

where:

$$c = \max_{i,j} |\langle m_i | n_j \rangle|^2 \quad (2.127)$$

Essentially, the above states that, given *any* quantum state, one cannot be certain of the outcome of two different measurements - there must always be some uncertainty in the system and the amount of uncertainty will depend on the “overlap” of the two bases, or, ultimately, how different they are.

Let’s consider some basic examples. Let’s say $M = N = Z$, that is, both measurement bases are the computational Z basis. In this case, $c = 1$ and, indeed, we can readily find quantum states where $H(Z) + H(Z) = 0$. Just consider $\rho = |0\rangle\langle 0|$. On the other hand, let’s say $M = Z$ and $N = X$. Now, $c = 1/2$ and so, for every state it should hold that $H(Z) + H(X) \geq 1$. We can check this at least for some states. Consider $\rho = |0\rangle\langle 0|$ as before. In this case, $H(Z) = 0$ (why?) but $H(X) = 1$ since an X basis measurement will result in a completely mixed state with maximal entropy. In this case, the bound above holds. You may check this for other states of your choosing; as it turns out, it always works.

Now, the above is interesting, however from a cryptographic standpoint, where there exists an adversary system, it is not as useful as it might be. However, there are many different entropic uncertainty relations (see, for instance, ?? for an excellent survey). One in particular was discovered in ?, and works for tripartite systems of the form ρ_{ABE} . Let’s assume Alice makes a measurement of her system in either the M or N basis (as before), leading to density operator ρ_{AMBE} or ρ_{ANBE} respectively (i.e., A_M is the new system representing Alice’s outcome in the M basis). This relation reads:

$$H(A_M|E)_\rho + H(A_N|B)_\rho \geq c, \quad (2.128)$$

where c is the same as in Equation 2.127. Now, we have a conditional entropy expression. This will be very useful in future chapters, when we analyze QKD systems and other cryptographic systems. In particular, we can allow Eve (E) to create a system, perhaps entangled with herself, and send part of it to Alice and part of it to Bob. So long as Alice and Bob have “small” uncertainty as determined by $H(A_N|B)$, then Eve will be very uncertain of Alice’s measurement outcome if she measures in an alternative basis. The greater the difference between the two basis (as computed by c), the more uncertain she will be.

The above relations are suitable for von Neumann entropy, however similar relations exist for quantum min entropy. Let ρ_{ABE} be a quantum state where, now, the A and B system consist of n registers each of dimension d (e.g., if dealing with qubits, as is often the case, $d = 2$ and so the dimension of A and B is 2^n). Let M and N be d -dimensional orthonormal bases with overlap function given by c in Equation 2.127. Then, it was shown in ? that:

$$H_\infty(A_N|E)_\rho + H_{\max}(A_M|B)_\rho \geq n \cdot c, \quad (2.129)$$

where $H_{\max}(A|B)$ is the so-called *max-entropy* defined by:

$$H_{\max}(A|B)_\rho = -H_\infty(A|C)_\rho, \quad (2.130)$$

where the above min entropy is evaluated for a *purification* ρ_{ABC} of ρ_{AB} . (To *purify* a mixed state ρ_{AB} means to add an additional system C that turns the entire joint state, ABC into a pure state.) We will not go into great detail on max entropy, here, but will recall necessary details, later, when required.

Importantly, the above uncertainty relation will be very useful in future chapters. In general, max entropy can be estimated by Alice and Bob directly; if it is small (i.e., Alice and Bob have a low degree of uncertainty between their outcomes), then the min entropy, between Alice and the Eve system, must be large.